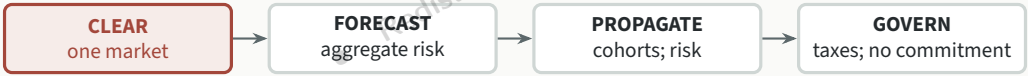
Session 4.1

Aiyagari, Solved Twice: A Fixed Point, Then a Law of Motion

The classical solve returns one stationary distribution. An actor–critic solve with the distribution as its state returns how every distribution evolves

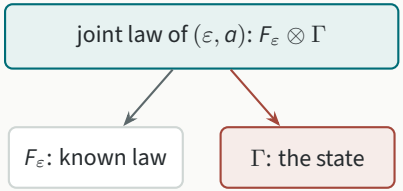


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The Economy



ε is redrawn every period, independently of a, so today's assets tell nothing about today's ε.

- $\log \varepsilon \sim N(0, 0.2^2)$, i.i.d.; $c + a' = (1+r)a + w\varepsilon l$, $a' \geq 0$.
- Utility $u(y) = y^{1-\sigma}/(1-\sigma)$ of the bundle $y = c - \psi_l \frac{l^{1+\theta_l}}{1+\theta_l}$: no wealth effect on hours, so $l(\varepsilon) = (w\varepsilon/\psi_l)^{1/\theta_l}$.
- $Y = K^\alpha L^{1-\alpha}$; r (net of δ) and w from marginal products; $K = \int a \, d\Gamma$.
- $\Gamma_{t+1} = \Phi(\Gamma_t)$. With i.i.d. ε the distribution of assets, Γ, is the whole state.
- **Prices depend on Γ, so the saving rule is $g(\varepsilon, a; \Gamma)$.**

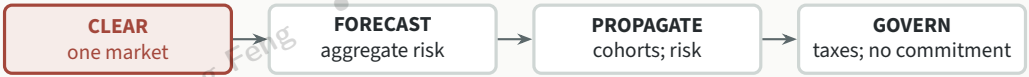
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Source: Feng, Han, Sargent & Zhu (September 2026), preliminary. $\beta = 0.95, \sigma = 2, \alpha = 0.34, \delta = 0.06, \theta_l = 0.32, \psi_l = 1.5$.

Session 4.1

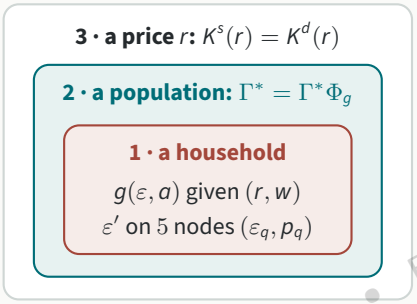
Stage 1A: The Classical Solve

A household, a population and a price: three fixed points, one inside the other



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Three Problems, Nested



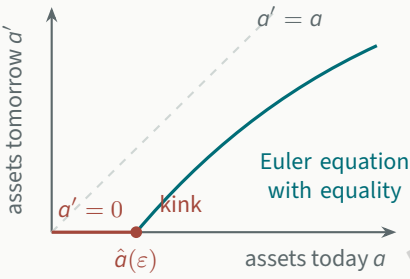
one evaluation of the outer residual: a full household solve and a full distribution solve

- **A household.** Given (r, w): $V(\varepsilon, a) = \max_{a' \geq 0} u(y) + \beta \sum_q p_q V(\varepsilon_q, a')$; with i.i.d. ε the expectation depends on a' alone.
- **A population.** The rule moves any Γ to $\Gamma' = \Gamma\Phi_g$; find the invariant $\Gamma^* = \Gamma^*\Phi_g$.
- **A price.** Find r at which $K^s(r) = \int a \, d\Gamma_r^*$ equals the firms' $K^d(r)$; w follows from r.
- $D(r) = K^d(r) - K^s(r)$: both terms must depend on r, or $D \equiv 0$, invisibly.
- **Each layer is a fixed point and contains the next: one evaluation of D costs a full dynamic program.**

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Source: Feng, Han, Sargent & Zhu (September 2026), preliminary, Stage 1A; Aiyagari, *QJE* (1994).

The Borrowing Limit Puts a Kink in the Rule



- With $a' \geq 0$: $u'(y) \geq \beta(1+r)\,\mathbb{E}\,u'(y')$, with equality when $a' > 0$.
- Below $\hat{a}(\varepsilon)$ the household would borrow; it cannot, so $a' = 0$: a kink in the rule.
- A kink defeats methods that assume smoothness; on a grid it is one comparison per node.
- Here 0.009% sit exactly at $a = 0$: i.i.d. risk needs only a small buffer. Persistence changes that (last section).
- **The constraint is handled by a comparison, not a solver.**

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Source: Feng, Han, Sargent & Zhu (September 2026), preliminary, Stage 1A.

The Household: the Endogenous Grid Method

- Grid on *tomorrow's* assets, Euler equation inverted, no search. In the bundle the budget is linear: $y + a' = (1+r)a + \phi(\varepsilon)$, with $\phi(\varepsilon) = w\varepsilon l(\varepsilon) - \psi_l \frac{l(\varepsilon)^{1+\theta_l}}{1+\theta_l}$, earnings net of the hours' disutility.

One sweep from the current guess g, on the fixed grid $a'_1 < \cdots < a'_n$

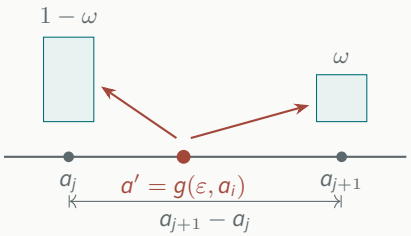
1. tomorrow's bundle at every (a'_j, ε_q): $y'_{jq} = (1+r)a'_j + \phi(\varepsilon_q) - g(\varepsilon_q, a'_j)$, no interpolation
2. $E_j = \sum_q p_q\, u'(y'_{jq})$: one number per a'_j, since ε' is independent of today
3. invert the Euler equation: $\tilde{y}_j = \left[\beta(1+r)E_j\right]^{-1/\sigma}$
4. today's assets that lead here, one per ε: $\tilde{a}_j(\varepsilon) = \left[\tilde{y}_j + a'_j - \phi(\varepsilon)\right]/(1+r)$
5. interpolate $(\tilde{a}_j(\varepsilon), a'_j)$ onto the fixed grid: the new $g(\varepsilon, a)$; $a' = 0$ for $a < \tilde{a}_1(\varepsilon)$

- No root-finding, no maximisation. Repeat until $\max|g_{\text{new}} - g| < 10^{-10}$; 1500 nodes, dense near the limit.
- **The constraint is where the interpolation runs out: a comparison, not a solver.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary, Stage 1A; Carroll, *Economics Letters* (2006).

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The Young Lottery



- The rule sends (ε, a_i) to $a' = g(\varepsilon, a_i)$, between two nodes: where does its mass go?
- Split it: $\omega = \frac{a'-a_j}{a_{j+1}-a_j}$ to a_{j+1}, $1-\omega$ to a_j. Rounding to one node would bias every moment.
- Then ε' is drawn with weights p_q. Mass is conserved and the conditional mean of a' is exact: Young (2010), often unnamed as "linear interpolation".
- **Deterministic. A simulated panel adds sampling noise to what we will later differentiate.**

Source: Young, *JEDC* (2010).

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From a Rule to a Population

- Lottery weights and ε draws define the **transition matrix** Φ_g (two nonzeros per (i, q)):

$$\Gamma' = \Gamma\,\Phi_g, \qquad (\Phi_g)_{ij} = \textstyle\sum_q p_q\,\omega_{iq\to j}, \qquad \omega_{iq\to j} = \text{share of } (\varepsilon_q, a_i)\text{'s mass sent to } j.$$

Three ways to $\Gamma^* = \Gamma^*\Phi_g$	**What it costs**
iterate $\Gamma_{t+1} = \Gamma_t\Phi_g$	simple; slow near a unit root
dense solve: $\Gamma^*(\Phi_g - I) = 0, \sum_i \Gamma^*_i = 1$ replacing one row	$O(n^3)$, exact; **Stage 1A**, $n = 1500$
the sparse eigenvector of $\Phi_g^\top$ at eigenvalue 1	fastest at scale

- Check the top of the grid: its top tenth holds 1.2×10^{-4} of the mass here; at 10^{-2} the grid would set K.
- **A histogram plus the lottery is the law of motion, as a matrix; Stage 1B rebuilds it every date.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary, Stage 1A; Young (2010).

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The Dense Solve: Trade One Redundant Equation for $\sum_i \Gamma_i^* = 1$

A toy chain, $n = 3$:

$$\Phi_g = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ 0 & \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

$(\Phi_g^\top - I)\,\Gamma^{*\top} = 0$, last row swapped:

$$\begin{bmatrix} -\frac{1}{2} & \frac{1}{4} & 0 \\ \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} \Gamma_1^* \\ \Gamma_2^* \\ \Gamma_3^* \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

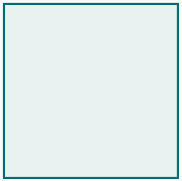
$$\Rightarrow \Gamma^* = \left(\tfrac{1}{4},\ \tfrac{1}{2},\ \tfrac{1}{4}\right)$$

- $(\Phi_g^\top - I)\,\Gamma^{*\top} = 0$: n equations, n unknowns.
- Rows of Φ_g sum to one, so the equations add up to $0 = 0$: one is redundant, and any multiple of Γ^* solves them.
- Swap it for $\sum_i \Gamma_i^* = 1$ (red): invertible, scale fixed.
- One LU factorisation: $\frac{2}{3}n^3$ operations on all n^2 entries, zeros included; $n^2 = 2{,}250{,}000$ here.
- **Exact in one shot; the cost grows like n^3 however sparse Φ_g is.**

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Source: Feng, Han, Sargent & Zhu (September 2026), preliminary, Stage 1A.

The Eigenvector Route: Multiply by $\Phi_g^\top$, Never Factor It



dense LU works on all n^2 entries



$\Phi_g^\top v$ touches at most $2Qn$

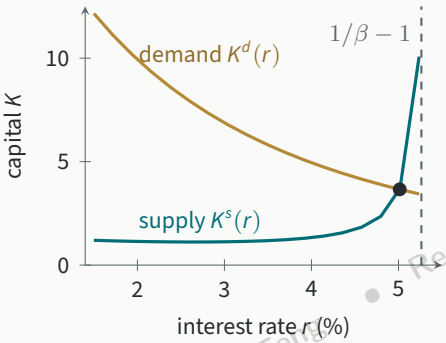
here $14{,}534$ against $2{,}250{,}000$

- Same equation, read as $\Phi_g^\top v = 1 \cdot v$: 1 is the largest eigenvalue; normalise after, $\Gamma^* = v / \sum_i v_i$.
- Plain iteration is the power method: error $\propto |\lambda_2|^t$. Toy: $\lambda_2 = \frac{1}{2}$. Here $\lambda_2 = 0.9972$: $t = \ln 10^{-10} / \ln \lambda_2 \approx 8{,}200$.
- Arnoldi uses the same products $v, \Phi_g^\top v, (\Phi_g^\top)^2 v, \ldots$ but combines them all: far fewer.
- **Same Γ^*; the cost is products over $2Qn$ nonzeros, so it wins as n grows.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary, Stage 1A transition matrix; λ_2 by Arnoldi (ARPACK).

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The Outer Loop: Bisection in a Proven Bracket, Not Newton



- The bracket is known: demand falls in r, supply diverges as $r \to 1/\beta - 1$. Prove the sign change, then halve to 10^{-8}.
- Each evaluation is one warm-started EGM and one distribution solve. Not Newton: supply is only piecewise smooth in r (the binding set jumps), so a finite difference is no derivative.
- $r^* = 5.014\%$ against $1/\beta - 1 = 5.263\%$: precautionary saving. $K^* = 3.6595$, $K/Y = 3.09$; 30 CPU-seconds in all.
- $\beta(1 + r^*) = 0.99763$**: so close to one, and it returns in every section that follows.**

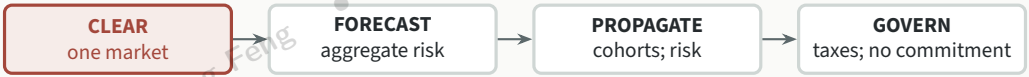
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Source: Feng, Han, Sargent & Zhu (September 2026), preliminary, Stage 1A; Aiyagari, *QJE* (1994).

Session 4.1

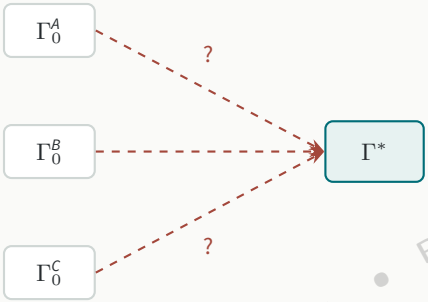
What the Classical Solve Cannot Answer

One fixed point is not a law of motion



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One Fixed Point, Not a Law of Motion



each dashed path is a separate problem: a sequence of prices, solved backward and forward

- Stage 1A says where the economy rests, not how it gets there.
- A transition from one Γ_0 is a price path and a shooting loop (Deck 4.3). A second Γ_0 is a second loop.
- Welfare along the way, or a policy that reacts to Γ, needs the answer at every Γ.
- **The unknown changes from a number, r^*, to a function of a distribution: $g(\varepsilon, a; \Gamma)$.**

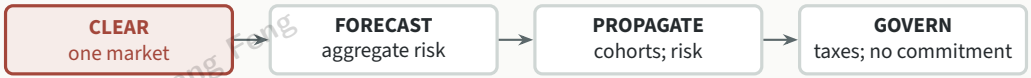
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Source: Feng, Han, Sargent & Zhu (September 2026), preliminary; Deck 4.3, *The Unknown Is a Path of Prices*.

Session 4.1

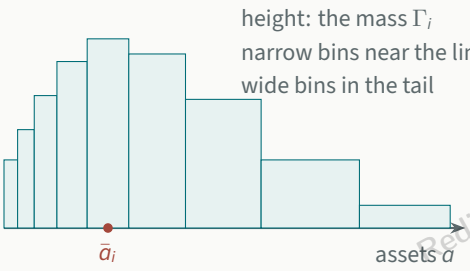
Characterising the Evolution of Γ

A histogram, a quadrature, and a transition matrix rebuilt inside the computational graph



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A Histogram and a Quadrature

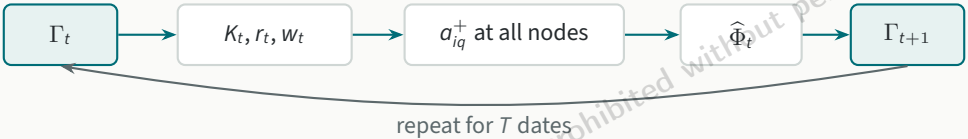


- $N = 140$ bins on $[0, 16]$ with midpoints $\bar{a}_i$; the state is $\Gamma = (\Gamma_1, \ldots, \Gamma_N)$, $\sum_i \Gamma_i = 1$, $K = \sum_i \Gamma_i \bar{a}_i$.
- ε: $Q = 5$ Gauss–Hermite nodes ε_q, weights p_q, as in Stage 1A. An expectation is a five-term sum.
- Grid chosen once from the benchmark: no bin above 1.6% of households, nothing off the top.
- **The discretisation is part of the model. Fix it for every method compared.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary.

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The Law of Motion, Rebuilt at Every Date



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From one distribution Γ_0, for $t = 0, \ldots, T-1$

1. $K_t = \sum_i \Gamma_{t,i} \bar{a}_i$; r_t, w_t from the firm, hours in closed form: no price loop
2. saving $a_{iq}^+ = g_\theta(\varepsilon_q, \bar{a}_i; \Gamma_t)$ at every node; consumption from the budget
3. build the $N \times N$ matrix $\widehat{\Phi}_t$ and set $\Gamma_{t+1} = \Gamma_t \widehat{\Phi}_t$

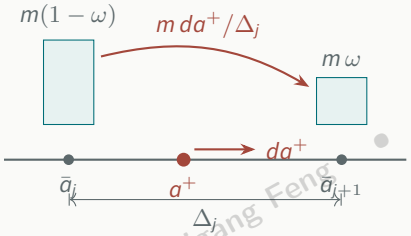
- **$\widehat{\Phi}_t$ depends on Γ_t and on the network, so it is rebuilt at every date of every rollout: computed, never simulated, inside the graph.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary.

Building $\widehat{\Phi}$, and Why This Kernel

Stage 1A's lottery, now from the network's a^+: for every source midpoint $\bar{a}_i$ and node ε_q

1. evaluate the saving network once: $a^+ = g_\theta(\varepsilon_q, \bar{a}_i; \Gamma)$
2. find the bracket $\bar{a}_j \le a^+ \le \bar{a}_{j+1}$, and the weight $\omega = (a^+ - \bar{a}_j)/(\bar{a}_{j+1} - \bar{a}_j)$
3. add $p_q(1-\omega)$ to $\widehat{\Phi}_{ij}$ and $p_q\,\omega$ to $\widehat{\Phi}_{i,j+1}$



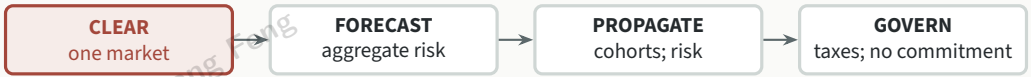
- **Exact in the mean**: the only split between neighbours with mean a^+.
- **Mass by construction**: each row of $\widehat{\Phi}$ sums to one.
- **Differentiable**: da^+ moves mass $m\,da^+/\Delta_j$ between neighbours, and autodiff chains it across dates.
- **The same lottery as Stage 1A; its derivative is what the rollout's gradient needs.**

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Session 4.1

Stage 1B: An Actor–Critic Solve

The saving rule is the actor, the household's value is the critic, and both read the histogram



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The Problem the Two Networks Solve

- At a distribution Γ, each household solves, taking Γ, prices and Γ_+ as given,

$$V(\varepsilon, a; \Gamma) = \max_{0 \le a' \le \bar{a}} \Big\{ u(y) + \beta \sum_{q'} p_{q'} \, V(\varepsilon_{q'}, a'; \Gamma_+) \Big\}, \qquad y = c - \psi_l \tfrac{l^{1+\theta_l}}{1+\theta_l},$$

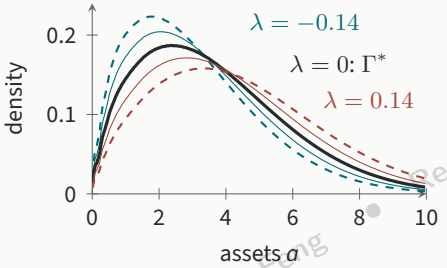
  with $c = (1 + r(\Gamma))\, a + w(\Gamma)\, \varepsilon\, l(\varepsilon; \Gamma) - a'$ and $\Gamma_+ = \Gamma\, \widehat{\Phi}(\Gamma; \theta)$.
- **Actor**: $a' = g_\theta(\varepsilon, a; \Gamma) = \max\{\tilde{a}_\theta, 0\}$, capped at $\bar{a}$: the **borrowing limit** is built into the rule, so $a' \ge 0$ holds at every node; where it binds, $c = (1 + r)a + w\varepsilon l$.
- **Critic**: $V_\psi \approx V$. Its Bellman target is computed from a trailing copy ψ^- of its weights, nudged after each critic step ($\psi^- \leftarrow 0.98\, \psi^- + 0.02\, \psi$), so the target holds still while ψ learns: the **target network**.
- **Both read the whole histogram: two hidden layers of 256 units on (ε, a) and the 140 masses.**

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Source: Feng, Han, Sargent & Zhu (September 2026), preliminary, Stage 1B.

Preparing the Batch: Whole Distributions Around Γ^*

Γ^* from Stage 1A → draw $\lambda \sim U[-0.14, 0.14]$ → $\Gamma_i^e \propto \Gamma_i^* \, e^{\lambda \bar{a}_i}$, renormalise → a batch of 128 Γ^e



- A training example is a whole distribution Γ^e, never a household; 128 fresh ones per iteration.
- Tilt Γ^* (five shown, $\lambda = 0, \pm 0.07, \pm 0.14$): $\lambda > 0$ moves mass to the rich, $\lambda < 0$ to the poor; K from 3.04 to 4.45.
- Each node $(\bar{a}_i, \varepsilon_q)$ of each Γ^e enters the loss: $N \times Q$ states per aggregate state.
- **Γ^e is today's distribution, $\Gamma_+^e = \Gamma^e \widehat{\Phi}(\Gamma^e)$ its successor; Γ^* is only the band's centre.**

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Source: Feng, Han, Sargent & Zhu (September 2026), preliminary, Stage 1B.

The Critic Step: Fit the Bellman Equation

- At every node $(\bar{a}_i, \varepsilon_q)$ of every Γ^e, with $a^+_{iq} = g_\theta(\varepsilon_q, \bar{a}_i; \Gamma^e)$ from the current actor, minimise over ψ

$$\mathcal{L}_V(\psi) = \underset{e,i,q}{\text{mean}} \left[V_\psi(\varepsilon_q, \bar{a}_i; \Gamma^e) - b^e_{iq} \right]^2,$$
$$b^e_{iq} = U^e_{iq} + \beta \sum_{q'} p_{q'} \, V_{\psi^-}\left(\varepsilon_{q'}, a^+_{iq}; \Gamma^e_+\right).$$

- U^e_{iq}: flow utility at a^+_{iq}. b is held fixed while ψ moves; its ε'-sum is exact, so no sampling noise.
- **One step: one Adam update of ψ on the whole batch (rate 3×10^{-3}), then $\psi^- \leftarrow (1 - 0.02)\, \psi^- + 0.02\, \psi$.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary, Stage 1B.

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The Actor Step: Maximise the Bellman Right-Hand Side

- Critic fixed, the actor maximises the Bellman right-hand side at every node of every Γ^e, with $a^+_{iq} = g_\theta(\varepsilon_q, \bar{a}_i; \Gamma^e)$:

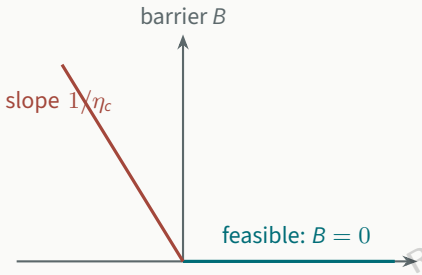
$$J(\theta) = \text{mean}_{e,i,q}\left[u(y^e_{iq}) + \beta\, C(a^+_{iq})\right], \qquad C(a') = \textstyle\sum_{q'} p_{q'}\, V_{\psi^-}(\varepsilon_{q'}, a'; \Gamma^e_+).$$

- One step: one Adam update of θ along $\nabla_\theta J = \text{mean}_{e,i,q}\left[\partial_{a^+}(u + \beta C)\right] \partial_\theta a^+_{iq}$, rate 5×10^{-5}.
- C's *value* is the critic's, its *slope* the envelope condition; so at an interior node $\partial_{a^+}(u + \beta C) = -u'(y) + \beta \sum_{q'} p_{q'}(1 + r_+)\, u'(y_{+,q'})$, the Euler residual, driven to zero.
- **At the limit the clamp has zero slope: no Euler equation at the constraint, none needed without a government; that household contributes nothing.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary, Stage 1B. $y_{+,q'}$: the bundle the current rule leaves at $(\varepsilon_{q'}, a'; \Gamma_+)$.

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Why the Barrier, and Why the Slope Is Replaced



- A proposed a' can make c or y negative, where $u(y)$ is undefined.
- $B = \left[\max(-c, 0) + \max(-y, 0)\right]/\eta_c$, $\eta_c = 10^{-4}$: a hinge, zero wherever the proposal is feasible.
- The actor maximises $J - \kappa B$, $\kappa = 1$; at the optimum $B = 0$.
- Why the envelope slope: at $\beta R = 0.99763$ the Euler terms nearly cancel, and a fitted V's slope is too noisy to resolve that.
- **One barrier for feasibility, one exact slope; neither changes what the actor maximises.**

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Source: Feng, Han, Sargent & Zhu (September 2026), preliminary, Stage 1B.

What Pins K: the Rollout Penalty

$$\mathcal{L}_g(\theta) = -J(\theta) + \kappa \operatorname*{mean}_{e,i,q} B^e_{iq}(\theta) + \lambda_K \frac{1}{T}\sum_{t=1}^{T}\left(K_t(\theta) - K^*\right)^2$$

- $-J$ is the actor's Bellman objective; κ mean B is the feasibility barrier of the last frame, $\kappa = 1$: zero wherever consumption is feasible, so it never moves the optimum.
- The **rollout penalty**: roll Γ^* forward $T = 100$ periods under the current rule, inside the graph ($\Gamma_{t+1} = \Gamma_t \widehat{\Phi}(\Gamma_t; \theta)$, $K_t = \sum_i \Gamma_{t,i}\bar{a}_i$), and penalise the capital path's squared distance from K^*, weight $\lambda_K = 1000$.
- Why: with $\beta R = 0.99763$ a one-signed bias in saving is amplified about $1/(1-\beta R) \approx 420$ times in the stationary K. A one-step residual cannot see it; a hundred dates can.
- 2 **Adam steps on $\mathcal{L}_g$ per iteration. The rollout, not a residual, pins K.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary, Stage 1B.

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The Algorithm

Stage 1B: the household's actor–critic, no government

1 initialise the actor by regression on EGM rules at constant prices, one per sampled K
2 pre-fit the critic to that actor, 6000 steps, so the actor cannot exploit a bad critic
3 draw 128 tilted distributions Γ^e
4 $\quad 8$ critic steps: each one Adam update of ψ on $\mathcal{L}_V$, then $\psi^- \leftarrow (1-0.02)\psi^- + 0.02\,\psi$
5 $\quad 2$ actor steps: each one Adam update of θ on $\mathcal{L}_g$
6 repeat 4000 times; at intervals, find the actor's own stationary Γ and score it

- A **step**: one gradient update of one network on the batch of 128 distributions.
- 8 critic steps for 2 actor steps: the critic's target moves whenever the rule moves, so it takes more and larger steps to stay fitted.
- **The actor's slope comes from the envelope condition; what the critic steps buy is a fitted $V_\psi(\varepsilon, a; \Gamma)$, the household value Deck 4.4 loads.**

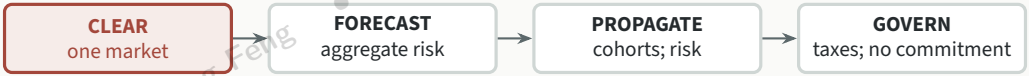
Source: Feng, Han, Sargent & Zhu (September 2026), preliminary.

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Session 4.1

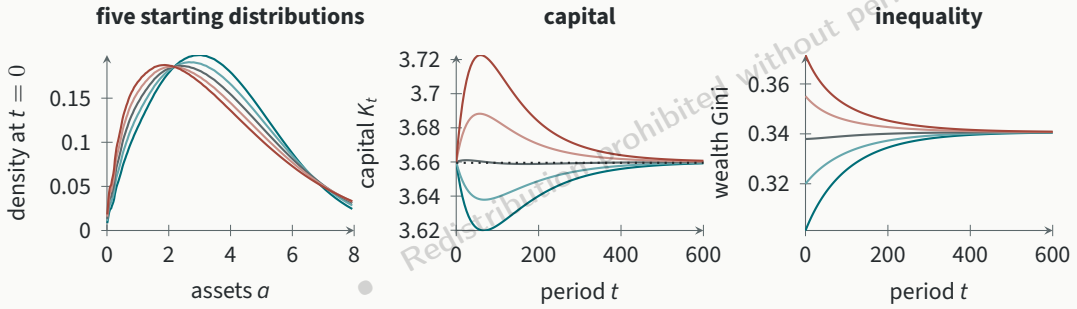
What It Returns, and Where It Stops

The distribution in motion, and why persistence changes the problem



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Same Capital, Different Inequality

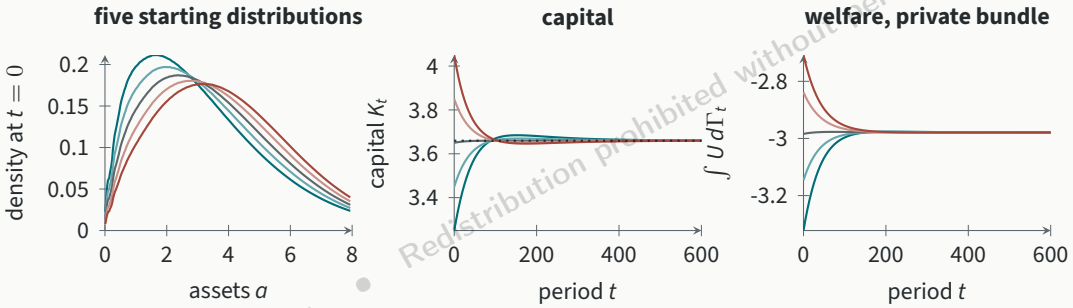


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- All start at $K_0 = 3.660$, Gini 0.301–0.372; all reach Gini 0.341, from below and above.
- K **leaves and returns (**3.620–3.723**). A state summarised by its mean would see nothing.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary, Stage 1B. Dotted: Stage 1A's K^*.

Same Spread, Different Capital



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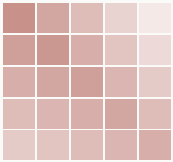
- K_0 from 3.25 to 4.05: K returns in about a hundred periods, the Gini far later.
- **Welfare along the path, from any start: what Stage 1A cannot return.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary, Stage 1B. Dotted: Stage 1A's K^*. No government, so no public good.

Where the Characterisation Gets Hard



N masses



$N \times n_\varepsilon$ masses

a and ε become correlated: the poor are poor because their ε was low

- Persistent ε: the joint law of (ε, a) no longer factorises. The state is a joint histogram, and $\widehat{\Phi}$ must carry the chain Π.
- With i.i.d. risk only 0.009% sit at the limit. In the lab's economy, with persistence 0.9, 2.9% do: runs of low ε walk them down to it.
- An aggregate shock z makes the map $\Phi(\Gamma, z, z')$ random, to be learned on every branch.
- **Each step multiplies the size of $\widehat{\Phi}$ and of every rollout in memory.**

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Source: Feng, Han, Sargent & Zhu (September 2026), preliminary; the lab, block A; Krusell & Smith, *JPE* (1998).

Bridge: When Φ Cannot Be Written Down, Simulate It

This deck. Γ is a histogram, Φ a matrix computed in the graph: exact, deterministic, differentiable.

persistence, aggregate risk

Next. Γ is a simulated panel, and the network learns from the states that the panel visits.

- Deep Equilibrium Nets and DeepHAM simulate households, and train where the simulation goes.
- The price: sampling noise, no derivative through a head count, coverage only where the policy goes.
- The gain: persistence and aggregate risk cost a longer simulation, not a bigger matrix.
- **Deck 4.2: three such methods. Deck 4.4: this histogram, with a government.**

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Source: Azinovic, Gaegauf & Scheidegger, *IER* (2022); Han, Yang & E (2026).

Session 4.2

Krusell–Smith, Solved Three Ways

Generalised moments, a histogram, a history of shocks: three ways to put a moving distribution into a network, and what each can and cannot see

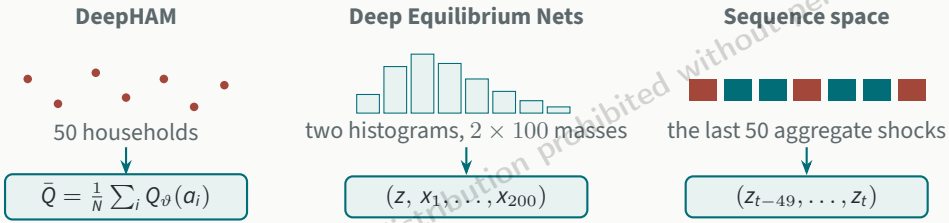


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Three Ways to Give a Network a Moving Distribution



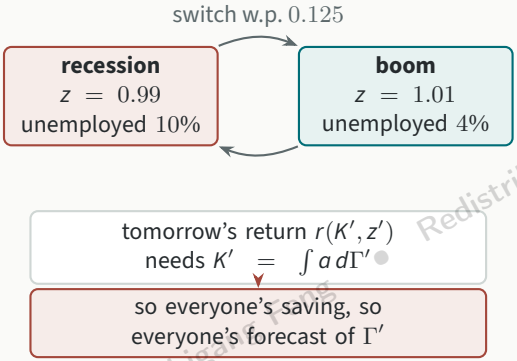
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- Krusell and Smith's economy: the household's state is $(a, \varepsilon; z, \Gamma)$, and Γ is a distribution.
- Each replaces Γ by something finite. For each: what the network reads, what it minimises, *where its training distributions come from*.
- **Then Deck 4.1's method, which writes down how Γ moves.**

Source: Han, Yang & E, *QE* (2026); Azinovic, Gaegauf & Scheidegger, *IER* (2022); Azinovic-Yang & Žemlička (2025).

The Economy, and Why the Household Must Forecast Γ

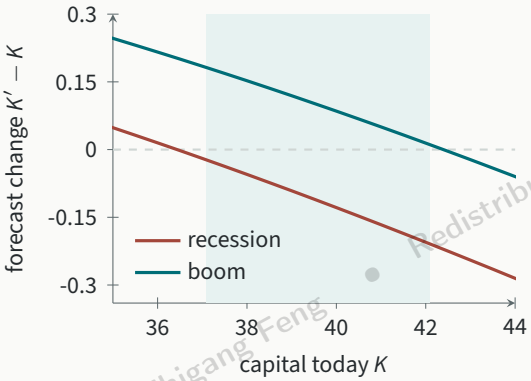


- $V(a, \varepsilon; z, \Gamma) = \max_{a' \geq 0} \ln c + \beta \, \mathbb{E} \, V(a', \varepsilon'; z', \Gamma')$, $c + a' = (1 - \delta + r)a + y(\varepsilon)$.
- r, w are marginal products at $K = \int a \, d\Gamma$; employment $\varepsilon \in \{0, 1\}$ moves with z.
- $\Gamma' = H(\Gamma, z, z')$**: the law of motion every household needs and nobody is given.**

Source: Krusell & Smith, *JPE* (1998); Den Haan, Judd & Juillard, *JEDC* (2010).

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The Classical Answer: Forecast With the Mean



- The bet: $\log K' = \kappa_0(z) + \kappa_1(z) \log K$. The state shrinks to (a, ε, K, z).
- Solve at the rule, push a histogram $3{,}000$ quarters, regress, update: 17 CPU-seconds.
- $R^2 > 0.99999$ one step ahead. Run alone for $2{,}000$ quarters: mean miss 0.054%, worst 0.187%.
- **The benchmark, and the idea each deep method generalises.**

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Source: Krusell & Smith (1998); Den Haan, *JEDC* (2010); the lab, block B. Quarterly, as published: $\alpha = 0.36$, $\beta = 0.99$, $\delta = 0.025$, $u = \ln c$, $z \in \{0.99, 1.01\}$.

Four Questions Every Deep Method Answers

	DeepHAM	Deep Equilibrium Nets	Sequence space
1. what stands for Γ	learned moments of a 50-household panel	a 100-point histogram per employment state	the last 50 aggregate shocks
2. what is minimised	minus lifetime utility over a rollout; a value regression	Euler and KKT residuals	a Fischer–Burmeister Euler residual
3. where training states come from	simulated panels, refreshed	16 simulated economies, carried forward	a replay buffer of simulated histories
4. what the gradient passes through	150 simulated periods of the panel	one period, at stored next states	one period; prices held fixed

- **Row 3: in all three, training distributions come from simulating the network's own policy.**

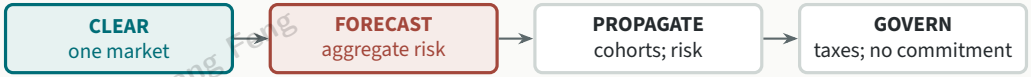
Source: DeepHAM's public repository; the Deep Equilibrium Nets continuum-of-agents code; the sequence-space tutorial.

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Session 4.2

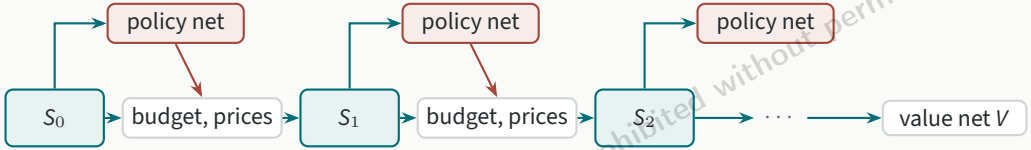
Method 1: DeepHAM

A network inside an average learns the moments that matter, and the policy is trained on simulated paths



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DeepHAM in One Picture: an Unrolled Panel



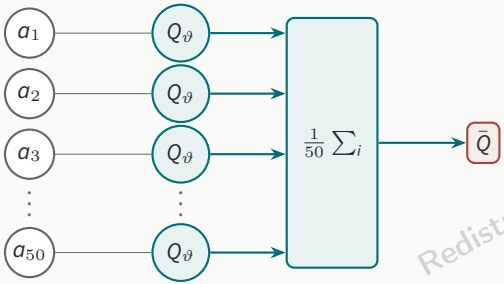
S_t: all 50 households' (a_i, ε_i) and z_t; prices from $K_t = \frac{1}{50}\sum_i a_i$

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- 384 economies run side by side, each a panel of $N = 50$ households.
- The actor simulates 150 periods forward from a stored cross-section inside the loss; a critic values what is left. Where the stored cross-sections come from: the training-loop pages.
- **Both networks read the panel through generalised moments.**

Source: Method 1 follows Han, Yang & E, "DeepHAM: A Global Solution Method for Heterogeneous Agent Models with Aggregate Shocks", *QE* 17(2), 2026 (arXiv 2112.14377), and its public repository (Figure 2, §3). At CPU scale in PyTorch: lab block D.

Generalised Moments: a Network Inside an Average



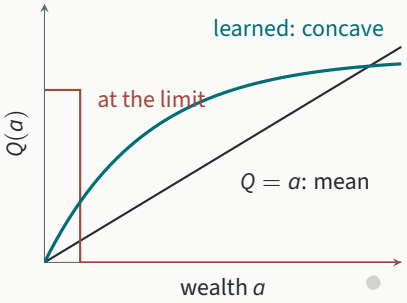
one network, shared by all:
2 layers of 12 tanh units

- $\bar{Q}_t = \frac{1}{N}\sum_i Q_\vartheta(a_t^i)$: one small network, applied to each household's wealth, then averaged.
- Household i's policy reads $(a_i, \varepsilon_i, z_t, \bar{Q}_t)$.
- ϑ is trained with the policy and value networks: the moment is whatever lowers the loss.
- **Krusell–Smith run: $J = 1$, one generalised moment of wealth.**

Source: The paper, §2.2; repository *util.py*; configurations *game_nn_n50_0fm1gm.json* ($J = 1$) and *game_nn_n50.json* (the mean), identical in everything else.

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Why an Average, and What It Nests

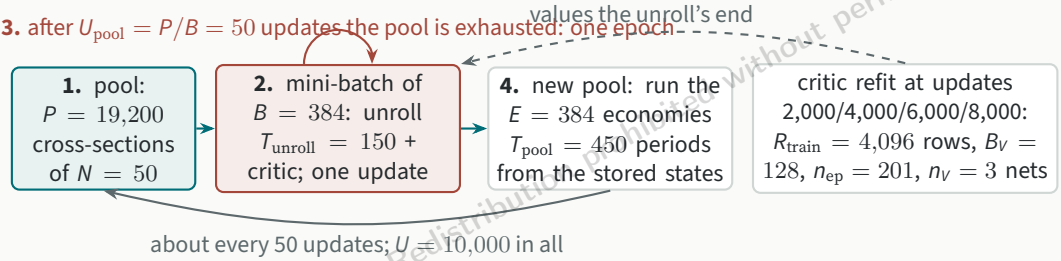


- **Permutation invariance**: renumbering households leaves $\sum_i Q(a_i)$ unchanged.
- $Q(a) = a$ gives the mean, Krusell and Smith's state; an indicator of the limit gives the hand-to-mouth share.
- Learned here: Q concave in wealth, value linear in $\bar{Q}$. Moving wealth from rich to poor raises $\bar{Q}$ at the same K.
- **Bellman error: the Krusell–Smith rule 0.0253; DeepHAM with the mean 0.0184; with one generalised moment 0.0151.**

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Source: The paper, §3.2, Table 2, Figure 3. Curves schematic.

DeepHAM's Training Loop in One Picture

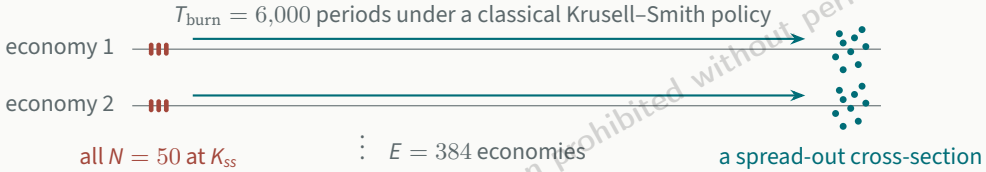


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- **1–2.** A pool of $P = E \times n_{\text{snap}} = 19{,}200$ cross-sections of $N = 50$; a mini-batch is $B = 384$ of them. **3–4.** After $U_{\text{pool}} = P/B = 50$ updates the pool is exhausted; a new pool from the stored initial states.
- **The $T_{\text{unroll}} = 150$-period unroll is inside the loss: it moves nothing in the pool.**

Source: Repository *train_KS.py*, *policy.py* (*train*, *sampler*), *dataset.py*; the Krusell–Smith configuration. Every symbol is collected on this method's last page.

1. Samples: 384 Economies of 50, Burned In



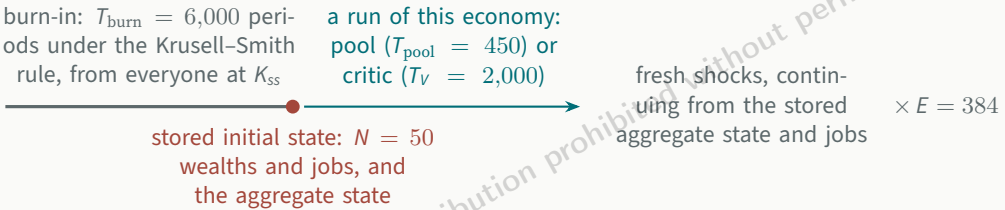
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- $E = 384$ economies side by side, each with $N = 50$ households and its own aggregate shocks.
- All start at K_{ss}; employment drawn at the aggregate state's unemployment rate.
- After $T_{\text{burn}} = 6{,}000$ periods, each economy's final cross-section is stored as its **initial state**.
- **A "distribution" is a sample of $N = 50$ households; the algorithm holds $E = 384$ of them.**

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Source: Repository *dataset.py* (*KSInitDataSet*), *simulation_KS.py*, the Krusell–Smith configuration.

1. Where Every Run Starts: the Stored Initial State

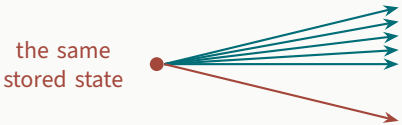


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- After the burn-in each of the $E = 384$ economies has a **stored initial state**: its $N = 50$ households' wealth and employment, and the aggregate state.
- Every later run of that economy, a pool run of $T_{\text{pool}} = 450$ or a critic run of $T_V = 2{,}000$, starts from its stored state, with fresh shocks.
- **A run starts from the stored state. What moves the stored state: the next two pages.**

Source: Repository *dataset.py* (*update_with_burn*, *update_from_simul*), *simulation_KS.py* (*simul_k*, *simul_shocks*).

1. Before the First Refit: Every Run Restarts From the Same State



pool runs, $T_{\text{pool}} = 450$ each, fresh shocks: about one per $U_{\text{pool}} = 50$ updates, about 40 before the first refit

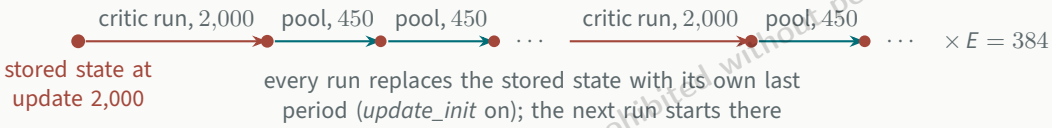
the first critic run, $T_V = 2{,}000$, before any update, under the constant consumption share 0.4

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- For updates 0–1,999 the stored states stay the burned-in ones: each pool run restarts from the same state, with new shocks.
- The first critic run, before any update, starts there too; its rows are the first critic fit's data.
- **About 40 pool runs from one state, with different shocks: nothing accumulates yet.**

Source: Repository *policy.py* (*train*: *update_init* False until the first refit), *dataset.py* (*get_policydataset*, *get_valuedataset*).

1. After the First Refit: Runs Join End to End

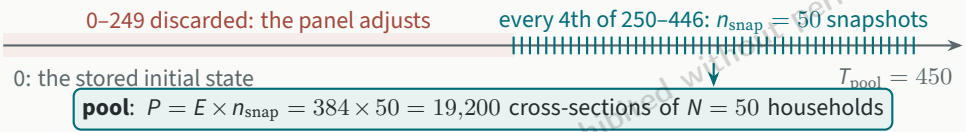


- At update $F = 2{,}000$ the critic run replaces each economy's stored state with its own last period, and so does every run after it, pool or critic.
- From then on each economy is one long simulation under the evolving rule: about 80,000 periods by update $U = 10{,}000$.
- **The same $E = 384$ economies feed the actor's pool and the critic's data; each datum is a cross-section of $N = 50$.**

Source: Repository *policy.py* (*train*: *update_init* set at the first refit, never reset), *dataset.py* (*update_from_simul*). Lengths not to scale.

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2–3. Actor: Build the Pool, Draw Mini-Batches, Exhaust It, Rebuild



One pool, then $U_{\text{pool}} = 50$ updates

1. build the pool: E economies run $T_{\text{pool}} = 450$ periods; keep every 4th of 250–446: $P = 19{,}200$
2. draw a mini-batch: $B = 384$ rows without replacement; fresh shocks for $T_{\text{unroll}} = 150$ periods
3. one Adam update of θ on the unroll loss, rate 4×10^{-4}
4. repeat 2–3; after $U_{\text{pool}} = P/B = 50$ updates every row is used once: *exhausted*, back to 1

- $U = 10{,}000$ **updates in all, one per mini-batch, about 200 pools. The pool supplies starting states, not targets.**

Source: Repository *dataset.py* (*get_policydataset*, *crazyshuffle*, *next_batch*) and *policy.py* (*sampler*, *train*); *epoch_resample* $= 0$ in the configuration.

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4. Three Clocks, and the Algorithm

clock	every	what is simulated, from where, and what it feeds
actor update	each mini-batch of B	$T_{\text{unroll}} = 150$ periods inside the loss, then discarded; one Adam step on θ
pool rebuild	$U_{\text{pool}} \approx 50$ updates	$T_{\text{pool}} = 450$ periods of all E from the stored states; the next 50 mini-batches
critic refit	$F = 2{,}000$ updates	$T_V = 2{,}000$ periods of all E from the stored states, then moved; $R = 4{,}608$ rows, $n_V = 3$ nets

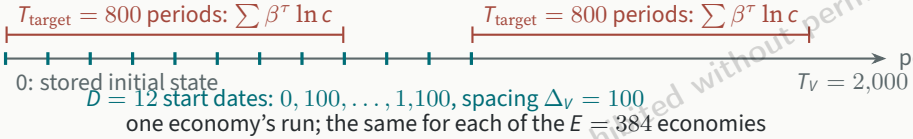
The algorithm: $U = 10{,}000$ actor updates

1 burn in the $E = 384$ economies for T_{burn}; at the constant share 0.4: critic run, fit, pool run
2 for $U = 10{,}000$ updates: a mini-batch of B, unroll T_{unroll}, one Adam step on θ
3 about every $U_{\text{pool}} = 50$ updates: rebuild the pool from the stored states
4 at updates 2,000/4,000/6,000/8,000: a critic run and a refit; the stored states move from now on

Source: Repository *train_KS.py*, *policy.py* (*train*, *sampler*), *dataset.py* (*update_from_simul*); *num_step*, *freq_update_v* in the configuration.

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The Critic's Data: One Economy's 2,000 Periods, 12 Dates; Times 384

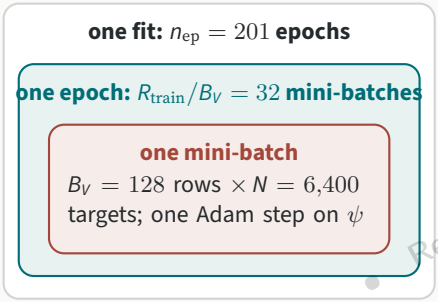


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- **Rows**: the figure is one economy's run; all $E = 384$ run $T_V = 2{,}000$ periods from their stored states. The cross-sections at the $D = 12$ dates are the rows: $R = D \times E = 4{,}608$.
- Why 12: a date needs $T_{\text{target}} = 800$ future periods ($1{,}100 + 800 < 1{,}999$); dates $\Delta_V = 100$ apart differ.
- **Samples**: a row keeps its $N = 50$ households, each with its own realised T_{target}-period target: $R \times N = 230{,}400$.
- $R_{\text{valid}} = 512$ **rows set aside to check the fit,** $R_{\text{train}} = 4{,}096$ **trained on. The critic's data are its own runs, not the actor's pool.**

Source: Repository *dataset.py* (*get_valuedataset*, *process_vdatadict*); *value_config* in the Krusell–Smith configuration.

The Critic's Fit: 32 Mini-Batches an Epoch, 201 Epochs, Three Nets



$U_{\text{fit}} = 201 \times 32 = 6{,}432$ updates per net; $\times\, n_V = 3$ nets, $\times\, n_{\text{fit}} = 5$ fits

- **Epoch**: one pass over the $R_{\text{train}} = 4{,}096$ rows in mini-batches of $B_V = 128$: 32 updates of ψ.
- **Fit**: $n_{\text{ep}} = 201$ passes over the *same* rows, reshuffled each pass: $U_{\text{fit}} = 6{,}432$ updates.
- **Three nets**: $n_V = 3$, same rows, different starts, averaged by the actor.
- **Fits**: once before training, then at 2,000, 4,000, 6,000, 8,000 ($n_{\text{fit}} = 5$); *frozen* between.
- **Tolerable: the critic carries only $\beta^{149} \approx 0.22$ of the objective, and a refit re-evaluates the current rule in full.**

Source: Repository *value.py* (*train*), *policy.py* (*train*, *freq_update_v*); *num_epoch*, *num_vnet*, *batch_size* in the configuration.

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5a. Inside One Actor Update: the Loss

- **Unroll**: simulate $T_{\text{unroll}} = 150$ periods forward *inside the loss*, from each S_0 in the mini-batch $\mathcal{B}$ of $B = 384$, so the gradient runs back through every period:

$$\max_{\theta} \frac{1}{B} \sum_{S_0 \in \mathcal{B}} \Big[\sum_{t=0}^{148} \beta^t \ln c_t^1 + \beta^{149} \bar{V}(S_{149}) \Big], \qquad c_t^i = \sigma\big(\mathcal{C}_\theta(a_i, \varepsilon_i, z_t, \bar{Q}_t)\big) \cdot \text{wealth}_t^i .$$

- $\bar{V}$ is the average of the $n_V = 3$ critics; $\beta^{149} \approx 0.22$ leaves the rest of the future to them. One Adam update of θ per mini-batch: $U = 10{,}000$ in all, rate 4×10^{-4}, constant.
- Only household 1 optimises; the other 49 use the same network with the gradient stopped, and set prices and $\bar{Q}$: **fictitious play**.
- **Not 384 cross-sections $\times$ 50 households per step: the reshuffle only makes household 1 a random member. Averaging all 50 would be a planner's problem.**

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Source: The paper, eq. (10), Algorithm 1; repository *policy.py* (*loss*, the "socialplanner" option).

5b. Inside One Critic Update: the Loss

- A mini-batch $\mathcal{B}$ is $B_V = 128$ training rows (s, e), a date and an economy; every household in each counts:

$$\mathcal{L}(\psi; \mathcal{B}) = \frac{1}{B_V N} \sum_{(s,e)\in\mathcal{B}} \sum_{i=1}^{N} \left[V_\psi\left(a^e_{i,s}, \varepsilon^e_{i,s}, z^e_s, \bar{Q}^{V,e}_s\right) - \widehat{V}^e_{i,s} \right]^2, \qquad \widehat{V}^e_{i,s} = \sum_{\tau=0}^{T_{\text{target}}-1} \beta^\tau \ln c^e_{i,s+\tau}.$$

- One Adam update of ψ per mini-batch, rate 10^{-4}: 32 per epoch, $U_{\text{fit}} = 6{,}432$ per fit, $n_{\text{fit}} = 5$ fits per net.
- A row keeps its $N = 50$ households together: each household's input includes $\bar{Q}^{V,e}_s$, the moment of its own cross-section.
- $\beta^{T_{\text{target}}} = \beta^{800} < 3.3 \times 10^{-4}$: the truncated sum stands in for the infinite one.
- **Monte Carlo evaluation, as in Session 3: each target is one realised future, and the regression averages the noise.**

Source: The paper, eq. (8)–(9); repository *dataset.py*, *value.py*.

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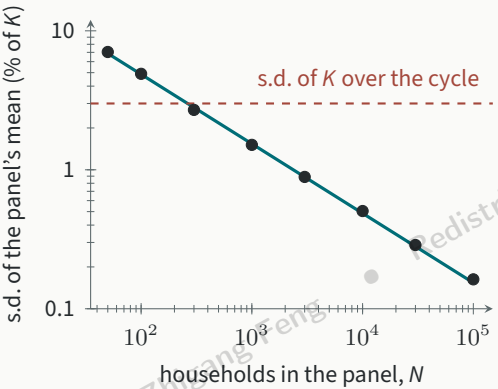
The Symbols

symbol	value	what it counts, and the formula
N, E	50, 384	households in one economy; economies run side by side
T_{burn}, T_{pool}	6,000, 450	burn-in periods, once; periods of one pool run
n_{snap}, P	50, 19,200	snapshots kept per pool run; the pool, $P = E \times n_{\text{snap}}$
B, T_{unroll}	384, 150	actor mini-batch ($B = E$ here); periods unrolled inside the loss
U, U_{pool}	10,000, 50	actor updates in all; updates one pool serves, $U_{\text{pool}} = P/B$
T_V, T_{target}, Δ_V	2,000, 800, 100	a critic run; a target's horizon ($\beta^{T_{\text{target}}} < 3.3 \times 10^{-4}$); date spacing
D, R	12, 4,608	start dates; rows of critic data, $R = D \times E$, each with N targets
R_{valid}, R_{train}	512, 4,096	rows set aside to check the fit; rows trained on
B_V, n_{ep}	128, 201	critic mini-batch in rows; epochs per fit; $R_{\text{train}}/B_V = 32$ an epoch
U_{fit}, n_V	6,432, 3	updates per fit per net, $n_{\text{ep}} \times 32$; nets averaged
F, n_{fit}, U_V	2,000, 5, 96,480	updates between refits; fits per net; critic updates, $n_{\text{fit}}\, n_V\, U_{\text{fit}}$

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Source: Repository configurations *game_nn_n50_0fm1gm.json* and *game_nn_n50.json* (identical but for the moment), *dataset.py*, *policy.py*, *value.py*.

What 50 Households Cost



- Wealth has s.d. 18.9 around $K = 38.8$: a panel's mean misses K by $18.9/\sqrt{N}$.
- $N = 50$: 2.67, 2.3 times the s.d. of K over the whole cycle.
- The paper treats $N = 50$ as the model, with similar results at 100 and 200.
- **A sampled state, and prices that inherit its noise.**

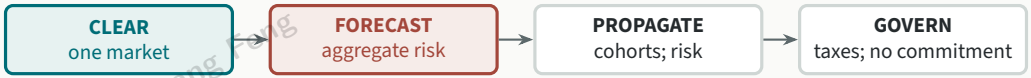
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Source: Lab block B, 400 panels per N. Quarterly, as published: $\alpha = 0.36, \beta = 0.99, \delta = 0.025, u = \ln c, z \in \{0.99, 1.01\}$. The paper, fn. 2.

Session 4.2

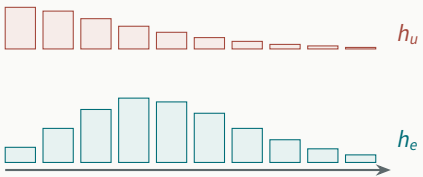
Method 2: Deep Equilibrium Nets

Keep the whole histogram, and ask only that the first-order conditions hold one period ahead



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The Same Economy, With a Histogram as the State



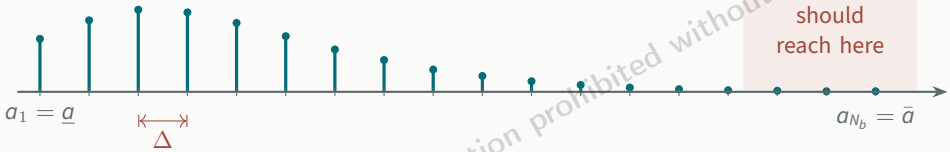
one asset grid of $N_b = 100$ points; a histogram of the unemployed and one of the employed

- State (z, Γ), with $\Gamma = (h_u, h_e)$: two histograms on one asset grid of $N_b = 100$ points, $2N_b = 200$ masses.
- Two, because employment is persistent: who is poor today predicts who is unemployed tomorrow, so assets alone do not determine next period.
- One network: $(\varepsilon, a; z, \Gamma) \mapsto (a', \lambda)$; prices from K.
- **Unlike DeepHAM: every mass is an input; no learned moment, no critics.**

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Source: Method 2 follows Azinovic, Gaegauf & Scheidegger, "Deep Equilibrium Nets", *IER* 63(4), 2022, and its public continuum-of-agents code, adapted to Krusell–Smith. Here: App. A.5; the code's state. A mean-capital version: lab block E.

Encoding the Distribution: Mass on a Grid of Points

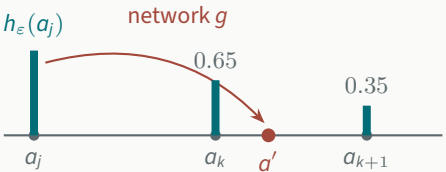


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- N_b *equidistant* points: the first is the limit $\underline{a}$; the last, $\bar{a}$, high enough that little mass reaches it. Savings above it are set to it.
- The code: $N_b = 100$ points on $[\underline{a}, \bar{a}] = [0, 20]$: $\Delta = \bar{a}/(N_b - 1) \approx 0.2$. A mass $h(a_j)$ sits *on* each point: a finite vector the network can read, and move exactly.
- **Unlike DeepHAM: the distribution is written on a grid, not sampled by 50 households.**

Source: The paper, following Reiter (2009) and Young (2010); the continuum-of-agents code (the asset grid).

From Today's Histogram to Tomorrow's: One Lottery Step



For each ε and grid point a_j

1. saving $a' = g(\varepsilon, a_j; z, \Gamma)$, set to a_{N_b} if above
2. a' lies between a_k and a_{k+1}
3. weights $w_k = 1 - |a_k - a'|/\Delta$, $w_{k+1} = 1 - w_k$
4. send $w_k h_\varepsilon(a_j)$ to a_k, $w_{k+1} h_\varepsilon(a_j)$ to a_{k+1}

$$h'_{\varepsilon'}(a_k) = \sum_{\varepsilon} \pi(\varepsilon' \mid \varepsilon; z, z') \sum_{j} h_\varepsilon(a_j)\, w_k\big(a'_{\varepsilon j}\big)$$

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- Then employment moves between the two histograms, with $\pi(\varepsilon' \mid \varepsilon; z, z')$: one Γ' for each z'. Mass is conserved; the mean of a' is exact.
- **Unlike DeepHAM: tomorrow's distribution is computed, $2N_b = 200$ network calls, not simulated household by household.**

Source: The continuum-of-agents code (the grid weights and the next-period histogram); Young, *JEDC* (2010).

The Equilibrium Conditions Are the Loss

- At a point $(\varepsilon, a; z, \Gamma)$, with $c = (1 - \delta + r)a + y(\varepsilon) - a'$ and c' from the same network at $(\varepsilon', a'; z', \Gamma')$:

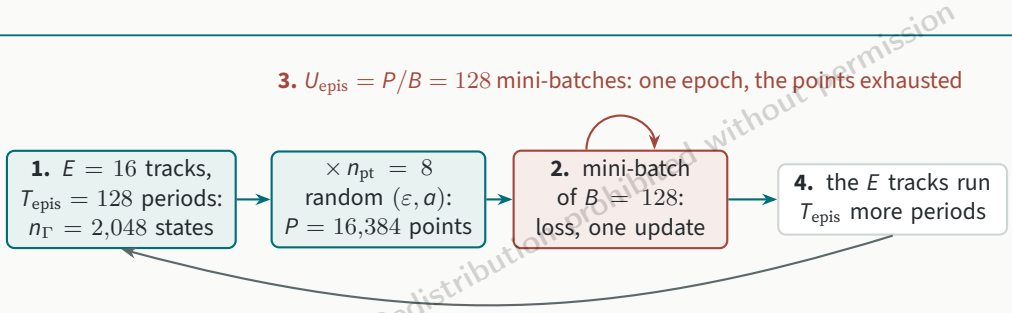
$$R_{EE} = \frac{\left[\beta\, \mathbb{E}\left[(1 - \delta + r')/c'\right] + \lambda\right]^{-1}}{c} - 1, \qquad R_{KKT} = a'\lambda.$$

- $\mathbb{E}$: an exact sum over the four (z', ε'). $R_{EE} = 0$ is the Euler equation; $R_{KKT} = 0$ lets $\lambda > 0$ only at the limit.
- $\Gamma'(z') = (h'_u, h'_e)$: the lottery step of the last page, one for each z'.
- **Unlike DeepHAM: first-order conditions one period ahead, not utility over a 150-period rollout.**

Source: The paper; the continuum-of-agents code (its Euler and KKT terms).

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Deep Equilibrium Nets' Training Loop in One Picture

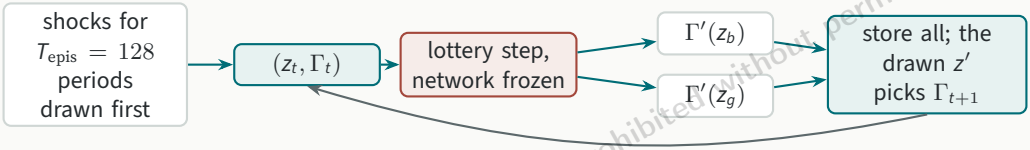


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- **1. Samples**: $n_\Gamma = 2{,}048$ states $\times\, n_{\text{pt}} = 8$ random (ε, a): $P = 16{,}384$ points.
- **2–4.** Mini-batches of $B = 128$: $U_{\text{epis}} = P/B = 128$ updates exhaust the points, one epoch.
- **No look-ahead in the loss: the successors were stored in the simulation.**

Source: The continuum-of-agents code (the episode loop, training data, batching). Symbols: this method's symbol page.

1. Samples: Simulated, and Stored With Their Successors



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- **Simulation for data**: $E = 16$ tracks, $T_{\text{epis}} = 128$ periods each per episode; the next page shows where they start and how they continue.
- Each period: one lottery step per z'; all successors stored; the pre-drawn shock picks the next state.
- **Inside the loss**: no simulation; $t + 1$ is a stored successor.
- **A one-period condition at states from a long simulation; DeepHAM simulates 150 periods inside its loss.**

Source: The continuum-of-agents code (shock simulation, the forward pass, the episode loop).

1. Samples: 16 Tracks, 128 Periods an Episode

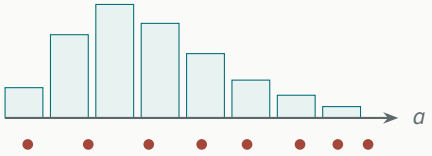


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- Episode 1: $E = 16$ copies of *one* state (wealth near zero), differing only by their shocks.
- Each episode: the code sets $n_\Gamma = 2{,}048$ states, so $T_{\text{epis}} = n_\Gamma / E = 128$ periods per track; at each dashed line a track continues from its own end state.
- **Not 16 hand-picked starts: $E = 16$ simulations of $n_{\text{epis}} T_{\text{epis}} \approx 8.3$ million periods each, in T_{epis}-period pieces, never restarted.**

Source: The continuum-of-agents code (starting states, shock simulation, the episode loop). The paths are schematic.

1. Samples: Each State Paired With 8 Evaluation Points



one stored (z, Γ), paired with $n_{\text{pt}} = 8$ random (ε, a): 8 evaluation points, not 8 agents

- **Data**: the $n_\Gamma = E\,T_{\text{epis}} = 2{,}048$ states (z, Γ) just simulated, with their stored successors.
- **Evaluation points**: each state (z, Γ) is paired with $n_{\text{pt}} = 8$ household states (ε, a), drawn at random: a uniform on $[\underline{a}, \bar{a}]$, ε a fair coin. $P = n_\Gamma\, n_{\text{pt}} = 2{,}048 \times 8 = 16{,}384$ points $(\varepsilon, a; z, \Gamma)$.

- (z, Γ) is not a starting point, and the 8 are not agents: the economy is a continuum. It is where the loss is evaluated, with its stored successors $\Gamma'(z')$.
- Why points: the network is pointwise, $(\varepsilon, a; z, \Gamma) \mapsto (a', \lambda)$; the Euler condition is a household's. Why uniform: the lottery uses the policy at all $2N_b = 200$ grid points.

Source: The continuum-of-agents code (training-data construction).

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2–3. Draw Mini-Batches, Exhaust the Points, Simulate Again

One episode's points, then $U_{\text{epis}} = 128$ updates

1 the points: $n_\Gamma = 2{,}048$ states $\times\, n_{\text{pt}} = 8$ random (ε, a): $P = 16{,}384$, shuffled
2 draw a mini-batch $\mathcal{B}$: $B = 128$ points, without replacement
3 one Adam update of θ on the loss of the next page, rate 10^{-5}
4 repeat 2–3; after $U_{\text{epis}} = P/B = 128$ updates every point is used once: *exhausted*, and discarded
5 the $E = 16$ tracks run $T_{\text{epis}} = 128$ more periods, network frozen: new points, back to 1

- **Epoch**: one pass over the points, U_{epis} updates: an episode is one epoch. $U_{\text{epis}} = T_{\text{epis}}$ because each simulated period yields $E\, n_{\text{pt}} = 128 = B$ points: one mini-batch.
- A point is used once and never again; DeepHAM's critic reuses its rows for 201 passes.
- **One Adam update of θ per mini-batch: $U = n_{\text{epis}}\, U_{\text{epis}} = 65{,}000 \times 128 \approx 8.3$ million in all. The points are evaluation points, not starting states.**

Source: The continuum-of-agents code (training data, shuffling and batching, the episode loop); *epochs_per_episode* $= 1$.

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5. Inside One Update: the Loss

- A mini-batch $\mathcal{B}$ is $B = 128$ evaluation points $(\varepsilon, a; z, \Gamma)$; each gives the Euler and KKT residuals R_{EE}, R_{KKT} defined above:

$$\mathcal{L}(\theta; \mathcal{B}) = \frac{1}{B} \sum_{(\varepsilon,a;z,\Gamma)\in\mathcal{B}} \Big[R_{EE}^2 + R_{KKT}^2\Big], \qquad \text{one Adam step at rate } 10^{-5}.$$

- The successors $\Gamma'(z')$ were stored during simulation: the network is re-evaluated at $(\varepsilon', a'; z', \Gamma')$, but how Γ moves is not differentiated.
- One period ahead only: the long simulation chooses the states; the loss never looks past $t + 1$.
- **Unlike DeepHAM: no rollout inside the loss and no critic; the gradient sees one period, at stored states.**

Source: The continuum-of-agents code (the loss and the training step).

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The Symbols

symbol	value	what it counts, and the formula
N_b, Δ	100, 0.2	grid points on $[\underline{a}, \bar{a}] = [0, 20]$; their spacing, $\bar{a}/(N_b - 1)$
E	16	tracks run side by side, identical at the start
n_Γ, T_{epis}	2,048, 128	states (z, Γ) simulated per episode (the code's setting); periods per track, n_Γ / E
n_{pt}, P	8, 16,384	random (ε, a) per state (2×4), not agents; evaluation points, $P = n_\Gamma \, n_{\text{pt}}$
B, U_{epis}	128, 128	mini-batch; updates per episode, $U_{\text{epis}} = P/B$ $(= T_{\text{epis}}$: $E \, n_{\text{pt}} = B)$
n_{epis}, U	65,000, 8,320,000	episodes; updates in all, $U = n_{\text{epis}} \, U_{\text{epis}}$
$n_{\text{epis}} \, T_{\text{epis}}$	8,320,000	periods each track is simulated over training

- All are settings of the public code: choices, not properties of the model.
- **One episode: simulate n_Γ states, pair them into P evaluation points, take U_{epis} steps, discard the points, move on.**

Source: The continuum-of-agents code: *num_tracks*, *len_episodes*, *num_id_per_shock*, *batch_size*, *num_episodes*, *nA*, *lr*.

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Two Clocks, and the Algorithm

clock	every	what is simulated, from where, and what it feeds
network update	each mini-batch of B points	nothing simulated: one Euler and one KKT residual per point, at its stored successors; one Adam step on θ
episode	$U_{\text{epis}} = 128$ updates	$T_{\text{epis}} = 128$ periods of all E tracks from their own end states, network frozen; $P = 16{,}384$ fresh points replace the old

The algorithm: $n_{\text{epis}} = 65{,}000$ episodes, $U \approx 8.3$ million updates

1 $E = 16$ tracks, identical at the start: a state (z, Γ), two N_b-point histograms
2 for each of $n_{\text{epis}} = 65{,}000$ episodes:
3 $T_{\text{epis}} = 128$ periods per track, network frozen; each period a lottery step per z', all stored
4 the $n_\Gamma = 2{,}048$ states $\times$ $n_{\text{pt}} = 8$ random (ε, a): P points; $U_{\text{epis}} = P/B = 128$ Adam steps

- **Two levers against DeepHAM: an exact state, not a sample; one period, not a long rollout.**

Source: The paper; the continuum-of-agents code (the episode loop; *num_episodes*, *epochs_per_episode*).

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Session 4.2

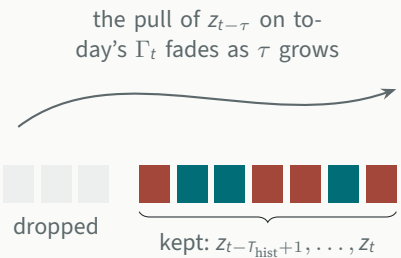
Method 3: The Sequence Space

Forget the distribution: the recent shocks already determine it



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The Distribution Is a Function of Past Shocks

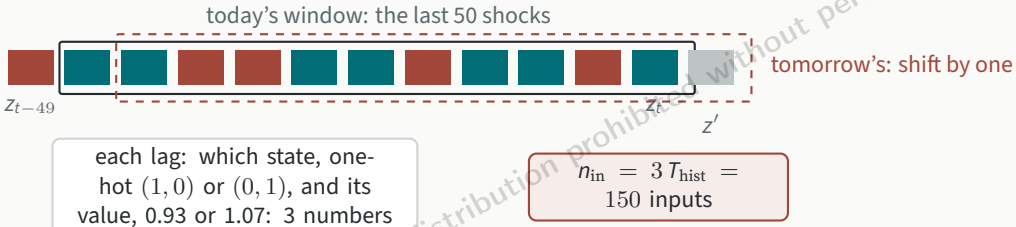


- Given the start and the policy, today's distribution is a function of all past shocks: $V(a, \varepsilon; \Gamma_t, z_t) = V(a, \varepsilon; z_0, \ldots, z_t)$.
- **Memory loss**: old shocks fade, so keep the last T_{hist}: $V \approx V(a, \varepsilon; z_{t-T_{\text{hist}}+1}, \ldots, z_t)$.
- The input is *exogenous*: the shock process sets its distribution, whatever the policy does.
- **Why it works: history $\to \Gamma$ is the law of motion iterated under one fixed policy. The tutorial keeps $T_{\text{hist}} = 50$.**

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Source: Method 3 follows Azinovic-Yang & Žemličkа, "Deep Learning in the Sequence Space", working paper, arXiv 2509.13623 (2025), with their slides (2026) and tutorial notebook; that notebook in PyTorch: the Session 4 lab, block F.

Encoding the Input: a Window of 50 Shocks

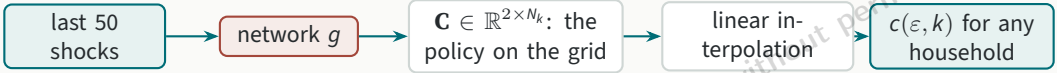


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- Stored as all $T_{\text{hist}} = 50$ one-hot pairs, then all 50 values: $n_{\text{in}} = 150$ numbers that stand in for the aggregate state, the distribution included. For tomorrow, drop the oldest shock and append z'.
- **Unlike DeepHAM: no household state and no moment enters; the input is exogenous, its distribution fixed by the shock process.**

Source: The tutorial notebook (history encoding and rolling windows).

Operator Learning: History In, a Whole Policy Out

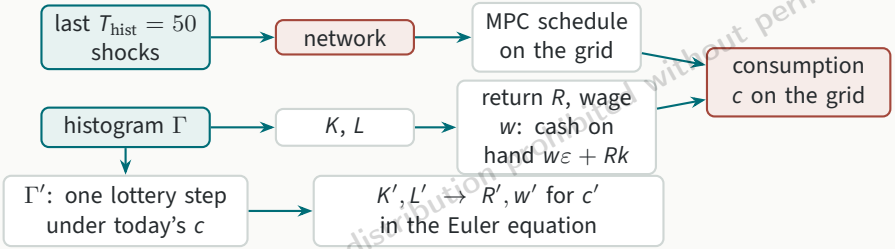


- **Operator learning** (Zhong, 2023; the paper extends it): write the policy as $f(X, x) = g(X)(x)$. The network g maps the aggregate input, here the history, to a whole *function* of the individual state $x = (\varepsilon, k)$.
- The function is stored as its values on a fixed grid, and read between grid points by linear interpolation. One call gives every household's policy.
- Shape is imposed at the grid points: the output is a starting share $\alpha(\varepsilon) < 1$ and falling MPCs, so c is increasing, concave and leaves $k' \geq 0$; linear interpolation keeps that shape between points.
- **Unlike DeepHAM, whose network is evaluated household by household: here the network's output is the policy function itself.**

Source: Zhong, Y. (2023), "Operator Learning in Macroeconomics", working paper; the tutorial notebook.

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Where the Distribution Still Enters: Through Prices

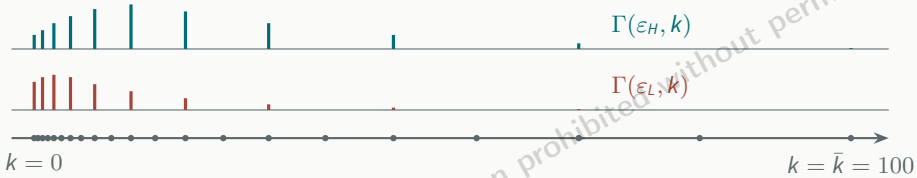


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- The network reads only shocks and gives the *shape* of consumption; the *level* needs today's prices, from Γ. Tomorrow's prices need Γ', one lottery step on.
- So every state is a pair (last $T_{\text{hist}} = 50$ shocks, Γ), with Γ simulated along those same shocks. Even a one-period loss needs a starting Γ.
- **The network reads shocks; the distribution sets the prices it is evaluated at.**

Source: The tutorial notebook (the network's forward pass takes the history and today's prices; the loss).

Why a Log-Spaced Grid



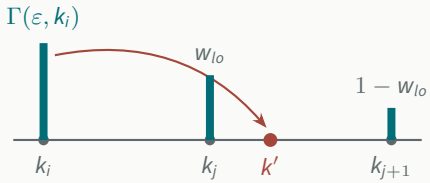
one grid for the policy's values and for the histogram (drawn schematically)

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- The grid is where the policy function lives. Linear interpolation errs where the function bends, and consumption bends most near the borrowing limit.
- So points are dense there: $N_k = 101$ points equally spaced in $\log(k + 0.5)$ on $[0, \bar{k}]$: $\Delta x \approx 0.053$, 0.03 apart near zero, 5 at the top. The paper suggests log-log spacing.
- **The histogram sits on the same grid: the lottery moves mass straight from the policy's own grid values.**

Source: The paper's section on operator learning; the tutorial notebook (capital grid).

The Same Lottery, on the Log Grid



For each ε and grid point k_i

1. $k' = w\varepsilon + Rk_i - c(\varepsilon, k_i)$, clipped to $[0, \bar{k}]$
2. bracket: $j = \lfloor (\log(k' + 0.5) - x_0)/\Delta x \rfloor$
3. $w_{lo} = (k_{j+1} - k')/(k_{j+1} - k_j)$
4. send $w_{lo}\Gamma(\varepsilon, k_i)$ to k_j, the rest to k_{j+1}

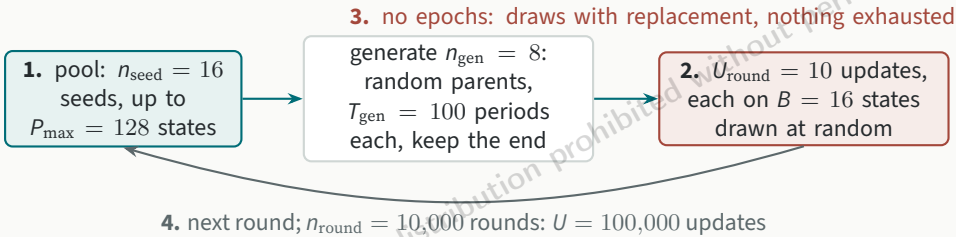
$$\Gamma' = \Pi_\varepsilon^\top \, \Gamma_{\text{after}}, \qquad K' = \sum_{\varepsilon,k} k\, \Gamma'(\varepsilon, k), \qquad L' = \sum_{\varepsilon,k} \varepsilon\, \Gamma'(\varepsilon, k).$$

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- Identical in spirit to Method 2's step; only the grid differs. In the tutorial, Π_ε does not depend on z.
- **What differs from Deep Equilibrium Nets is who reads Γ': there the network does; here only K' and L' do.**

Source: The tutorial notebook (the distribution step); Young, *JEDC* (2010).

The Sequence Space's Training Loop in One Picture



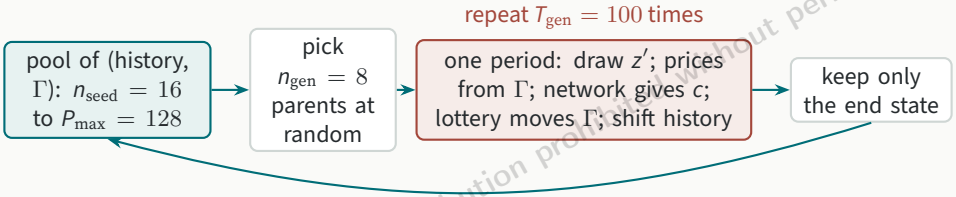
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- **1. Samples**: $n_{\text{gen}} = 8$ states a round, run $T_{\text{gen}} = 100$ periods from random parents; pool $\leq P_{\max} = 128$.
- **2–4.** $B = 16$ drawn with replacement per update; $U_{\text{round}} = 10$ a round, $n_{\text{round}} = 10{,}000$ rounds: $U = 100{,}000$.
- **The loss looks one period ahead (both z'): it moves nothing in the pool.**

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Source: The tutorial notebook (buffer, parent draws, training loop). Every symbol is collected on this method's symbol page.

1. Samples: One Round Adds 8 States

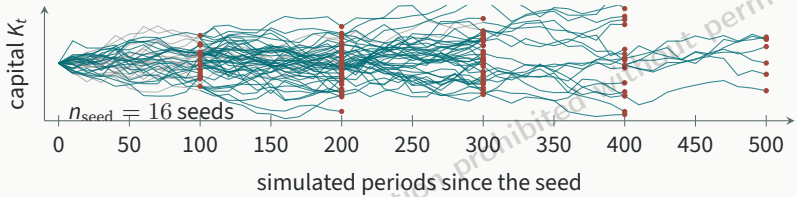


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- **Pool**: $n_{\text{seed}} = 16$ seeds (one all-low history, Γ_0), $+ n_{\text{gen}}$ a round: 16, 24, …, full at $P_{\max} = 128$ from round $(P_{\max} - n_{\text{seed}})/n_{\text{gen}} = 14$; then the oldest n_{gen} leave.
- **Parents**: a run starts at a parent drawn at random, *with replacement*, from the pool as it stands; a parent can be continued twice, and the seeds are the first parents.
- **Unlike DeepHAM:** $P_{\max} = 128$ **whole economies in all, renewed** $n_{\text{gen}} = 8$ **a round.**

Source: The tutorial notebook (seed, continuation, pool update).

1. Samples: a Branching Pool

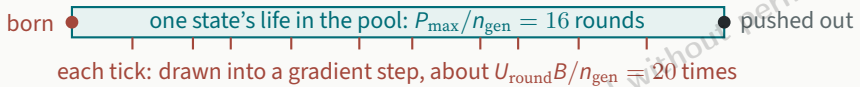


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- Each round: $n_{\text{gen}} = 8$ parents drawn with replacement; a child is its parent run $T_{\text{gen}} = 100$ periods on fresh shocks. Red: the pool after 20 rounds; grey: gone.
- **Depth**: a lineage gains T_{gen} a generation, and a parent's age averages 8.5 rounds (uniform over $P_{\max}/n_{\text{gen}} = 16$); after $n_{\text{round}} = 10{,}000$ rounds: $n_{\text{round}} T_{\text{gen}}/8.5 \approx 120{,}000$ periods.
- **Method 2 runs $E = 16$ fixed tracks; here tracks split and die out: a tree's tips.**

Source: The tutorial notebook (seeding, parent draws, continuation, pool update); the depth: a replay of that rule. Paths schematic.

2–4. Mini-Batches Drawn at Random; No Epochs; 10,000 Rounds



- **Mini-batch**: each update draws $B = 16$ states at random, *with replacement*, from the pool as it stands.
- **No epochs**: nothing is used up in order, nothing exhausted; a state stays $P_{\max}/n_{\text{gen}} = 16$ rounds and is drawn about $U_{\text{round}}B/n_{\text{gen}} = 20$ times (above).
- **Rounds**: $n_{\text{round}} = 10{,}000$: $n_{\text{gen}}n_{\text{round}} = 80{,}000$ states from $80{,}000 \times T_{\text{gen}} = 8$ million simulated periods, $P_{\max} = 128$ held at a time; $UB = 1.6$ million state-draws.
- **One Adam update of θ per mini-batch, $U_{\text{round}} = 10$ a round: $U = U_{\text{round}}\, n_{\text{round}} = 100{,}000$ in all, rate 5×10^{-5}, clipped at 1. Every state was simulated under a recent network.**

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Source: The tutorial notebook (training loop). The paper instead moves 4,096 pairs one period per episode.

Two Clocks, and the Algorithm

clock	every	what is simulated, from where, and what it feeds
network update	each mini-batch of $B = 16$ states	one period ahead (both z'), discarded; prices and Γ' fixed; one Adam step on θ
round	$U_{\text{round}} = 10$ updates	$T_{\text{gen}} = 100$ periods from $n_{\text{gen}} = 8$ parents drawn from the pool, network frozen; end states join, the oldest leave

The algorithm: $U = U_{\text{round}}\, n_{\text{round}} = 100{,}000$ **updates**

1 seed the pool: $n_{\text{seed}} = 16$ copies of (an all-low history, Γ_0)
2 for $n_{\text{round}} = 10{,}000$ rounds:
3 draw $n_{\text{gen}} = 8$ parents with replacement, run each $T_{\text{gen}} = 100$ periods, keep the end states
4 $U_{\text{round}} = 10$ times: draw $B = 16$ states with replacement, one Adam step on the FB loss

- **No separate burn-in, no epoch, nothing exhausted: the pool is a moving window of the last $P_{\max}/n_{\text{gen}} = 16$ rounds; DeepHAM burns in and exhausts each pool.**

Source: The tutorial notebook (training loop); *N_EPOCHS*: it calls a round an epoch.

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5. Inside One Update, Line by Line

One Adam step, of $U = 100{,}000$

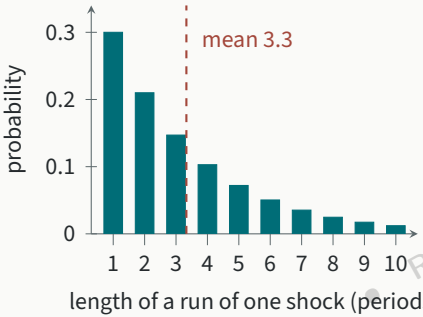
1. draw $B = 16$ states (history, Γ) at random, with replacement, from the pool
2. for each: R, w from Γ; one call $\mathbf{C} = g(\text{history}; R, w) \in \mathbb{R}^{2\times N_k}$; $k'_{ij} = w\varepsilon_i + Rk_j - C_{ij}$
3. Γ' = one lottery step of Γ under c; then K', L', held fixed
4. for each z': $\mathbf{C}' = g(\text{shifted history}; R', w')$, read at each k'_{ij} by interpolation
5. $q = \sum_{z',\varepsilon'} \pi\, \beta R'/c'$; then the Euler gap $e = 1/(qc) - 1$, $s = k'/c$, $\mathrm{FB}(e, s)$
6. loss = mean of FB^2 over $B \times 2N_k = 16 \times 202$; one Adam step, rate 5×10^{-5}, clipped at 1

- One network, one call per state: its output is the whole consumption function on the grid, $c(\varepsilon_i, k_j) = C_{ij}$; not a different network per node.
- The gradient reaches the network through $\mathbf{C}$ and $\mathbf{C}'$ only; prices, Γ and Γ' are held fixed.
- **Unlike DeepHAM: one period, every grid node of every state, no rollout and no critic.**

Source: The tutorial notebook (the loss and the training step).

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Coverage (1): The Draws Follow the True Shock Process

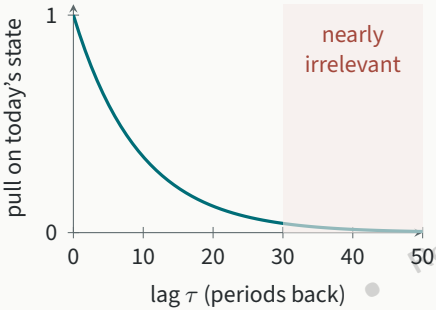


- All $2^{T_{\text{hist}}} = 2^{50} \approx 10^{15}$ histories are possible, and none is ruled out.
- But each new shock is drawn from the true chain: stay with probability 0.7, switch with 0.3. Training therefore meets histories as often as the economy does.
- Runs last $1/0.3 \approx 3.3$ periods on average, so likely histories are a small, structured part of the space.
- **Unlike DeepHAM and Method 2, the history's distribution does not depend on the network; only Γ does.**

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Source: The tutorial notebook (shock process and training loop).

Coverage (2): Old Shocks Matter Little, and the Distribution Comes Along



- **Memory loss**: an old shock's pull on today's state decays geometrically (in the paper's one-sector example, at rate $\max\{|\rho|, \alpha\}$). Histories that differ only far back need nearly the same answer.
- Γ needs no coverage of its own: it is whatever the simulated history produced.

- **Caveats:** each Γ in the pool was simulated under an older network; coverage is checked on simulated paths, not guaranteed; a pool of $P_{\max} = 128$ is a teaching setting.

Source: The tutorial notebook; the paper's argument that a truncated history is approximately sufficient. The decay curve is schematic.

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The Symbols

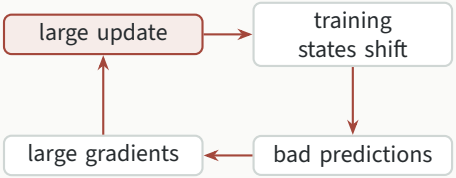
symbol	value	what it counts, and the formula
T_{hist}, n_{in}	50, 150	shocks in the window (the paper's T); inputs, $3\,T_{\text{hist}}$
N_k, $\bar{k}$	101, 100	grid points, equally spaced in $\log(k+0.5)$ on $[0,\bar{k}]$
Δx	0.053	that spacing, $[\log(\bar{k}+0.5)-\log 0.5]/(N_k-1)$; $2N_k = 202$ nodes (ε, k)
n_{seed}, $P_{\max}$	16, 128	seeds; pool capacity, full from round $(P_{\max}-n_{\text{seed}})/n_{\text{gen}} = 14$
n_{gen}, T_{gen}	8, 100	states generated per round; periods each is run
B, U_{round}	16, 10	mini-batch, drawn with replacement; updates per round
n_{round}, U	10,000, 100,000	rounds; updates in all, $U = U_{\text{round}}\, n_{\text{round}}$
$P_{\max}/n_{\text{gen}}$	16	rounds a state stays in the pool
$U_{\text{round}}B/n_{\text{gen}}$	20	times a state is drawn while it is there
$n_{\text{gen}}n_{\text{round}}$, UB	80,000, 1.6 million	states generated; state-draws

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- R, w, g: the notebook's gross return, wage and network; its μ is the distribution, Γ here.

Source: The tutorial notebook's settings (*N_AGG*, *BUFFER_CAP*, *FB_BATCH*, *FB_STEPS*, *N_EPOCHS*: it calls a round an epoch).

What the Exogenous Domain Buys



state-space training can feed on itself; a history's distribution does not move with the policy

- Reported: stable training across learning rates and seeds, where the state-space version struggles.
- Tutorial: $n_{\text{round}} = 10{,}000$ rounds in 191 CPU-seconds; mean $|\text{FB}| = 1.6 \times 10^{-4}$ over the pool's $P_{\max} = 128$ states. Its economy: annual; ε independent of z.
- The cost: the policy cannot be asked about a distribution no history produced.
- **Unlike the other two: the inputs do not shift when the policy does.**

Source: Stability: their slides (2026), pp. 11, 23–24. Time and $|\text{FB}|$: the tutorial's stored output.

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Three Pipelines, the Same Letters

	DeepHAM	Deep Equilibrium Nets	Sequence space
simulated	$E = 384$ economies, $N = 50$	$E = 16$ histogram tracks	$\leq P_{\max} = 128$ pool states
a run starts from	the stored state, burned in; moved after the first refit	the track's own end state	a parent drawn from the pool
periods per run	$T_{\text{pool}} = 450$	$T_{\text{epis}} = 128$	$T_{\text{gen}} = 100$
mini-batch B of P	384 of $P = 19{,}200$	128 of $P = 16{,}384$	16 of $P_{\max} = 128$
updates per dataset; datasets	$U_{\text{pool}} = 50$, exhausted; about $U/U_{\text{pool}} = 200$ pools	$U_{\text{epis}} = 128$, exhausted; $n_{\text{epis}} = 65{,}000$ episodes	$U_{\text{round}} = 10$, never exhausted; $n_{\text{round}} = 10{,}000$ rounds
updates in all, U	10,000 (critic 96,480)	8,320,000	100,000
look-ahead	$T_{\text{unroll}} = 150$, simulated	one period, stored	one period, computed

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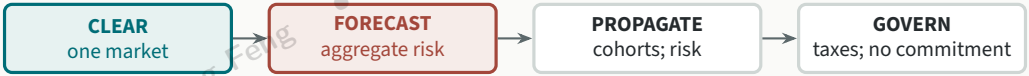
- **Same letters, three values: what differs is what a datum is and how long it lives.**

Source: The three symbol pages of this deck; the opener's questions 3 and 4, in symbols.

Session 4.2

Three Methods Against a Law of Motion

What each method moves forward, and what a government's problem needs



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How Far Forward Each Method Moves the Distribution

	what moves Γ	horizon in the loss	gradient through the move?
DeepHAM	a panel of $N = 50$, simulated	$T_{\text{unroll}} = 150$ periods	yes, through the sampled panel
Deep Equilibrium Nets	Young's lottery on a histogram	one period	no: successors are stored data
Sequence space	Young's lottery, beside the history	one period	no: prices and K' held fixed
Feng, Han, Sargent & Zhu (2026)	the lottery matrix $\widehat{\Phi}(\mathbf{x};\theta)$, in the graph	T: 100; 30 with a government	yes, through every $\widehat{\Phi}_t$

- **Methods 2 and 3 use the one-period lottery only to generate data; Feng, Han, Sargent & Zhu (2026) differentiate the whole path, DeepHAM through a sample.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary; the three sources' code, as on the previous pages.

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Argument 1 (a): What a Markov Government Has to Compute

- Each period the government picks a tax τ; households respond with g_τ; successors follow the rule $\tau(\cdot)$:

$$V^G(\Gamma, z) = \max_{\tau} \Big\{ W(\Gamma, z, \tau) + \beta\, \mathbb{E}\big[V^G(\Gamma', z') \mid z \big] \Big\}, \qquad \Gamma' = H\big(\Gamma, z, z';\, g_\tau\big),$$

  and in equilibrium the maximiser is the rule itself: $\tau(\Gamma, z)$.
- Its first-order condition:

$$\frac{\partial W}{\partial \tau} + \beta\, \mathbb{E}\Big[\frac{\partial V^G}{\partial \Gamma'} \cdot \frac{\partial H}{\partial g} \cdot \frac{\partial g_\tau}{\partial \tau} \Big] = 0.$$

- H is the law of motion, not a histogram: it maps today's distribution to tomorrow's. The second term: τ changes saving g_τ, which moves Γ' through H; with $\beta R \approx 1$ it compounds.
- **Needed: a rule that reads Γ, households that respond to τ, and $d\Gamma'/d\tau$ through H.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary; Klein, Krusell & Ríos-Rull, *RES* (2008).

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Argument 1 (b): Which Method Supplies What

	rule reads Γ	**households respond to τ**	$d\Gamma'/d\tau$	**successor governments**
DeepHAM	through a moment of 50	tax fixed; a planner option picks *c*	through a sampled panel	no
Deep Eq. Nets	yes, the histogram	not in the code	no: Γ' is stored data	no
Sequence space	no: shocks only	not in the code	no: prices held fixed	no
Feng, Han, Sargent & Zhu (2026)	yes, the histogram	yes: implementability	yes: through $\widehat{\Phi}$, every date	yes: a perceived rule

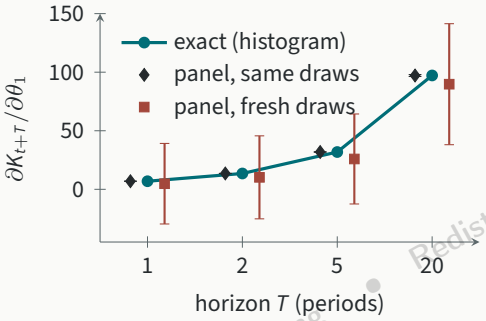
Feng, Han, Sargent & Zhu (2026) so far without aggregate risk: adding *z* is open.

- The sequence space fails at the root: history $\rightarrow \Gamma$ holds under *one fixed* policy; a deviating government breaks it. **Only a law of motion in the graph supplies the second term of the government's condition.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary; Han, Yang & E, arXiv 2112.14377, §2.4, §5; the three sources' code.

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Argument 2 (a): A Derivative From a Panel, and From the Histogram

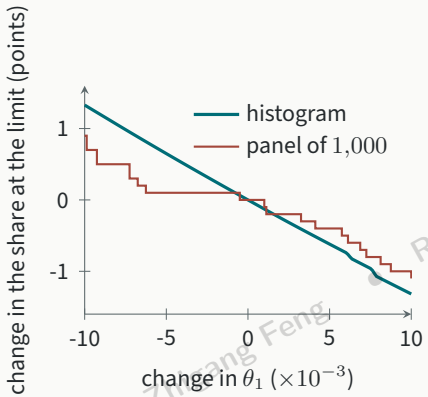


- **The test**: households save $a' = \theta_0 + \theta_1$ cash, clipped at the limit. How far does capital T periods on move per unit of θ_1?
- **Exact**, through the histogram: 6.90 at $T = 1$.
- **A panel** of $10{,}000$: simulate at $\theta_1 \pm 0.001$ and difference. Fresh draws: 4.8 ± 34.4; same draws: 6.90 ± 0.05. Mean $\pm$ one s.d. over 20 repetitions, as the bars.
- **For capital a panel works if its draws are fixed, as in DeepHAM's gradient.**

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Source: Lab block B: $\theta_1 = 0.938$, a linear rule fitted to the solved policy; the lab's annual economy.

Argument 2 (b): Where a Panel Breaks: Head Counts and Clamps

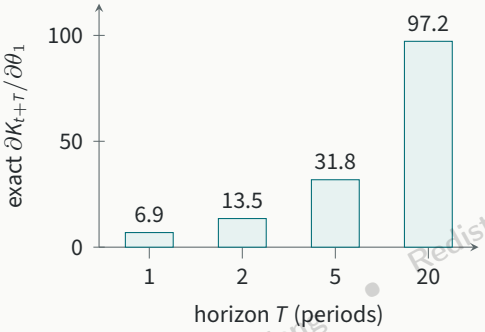


- Same test, now the share at the borrowing limit after 20 periods; one household is 0.1 point.
- A step's derivative is zero wherever it exists. At $T = 2$: exact -0.13; panel 0.00 ± 0.00.
- A clamped household ignores θ_1: a panel gradient misses who is pushed onto the limit.
- **Only the lottery in the graph, moving mass $h\,da'/\Delta$, gives the whole derivative.**

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Source: Lab block B's test swept over θ_1, same draws at every slope; the lab's annual economy. Feng, Han, Sargent & Zhu (September 2026), preliminary.

Argument 2 (c): Why the Derivative Must Look Many Periods Ahead



- At $T = 20$ the effect is 14 times its one-period size, and still rising; five periods capture 33% of it.
- With $\beta R \approx 1$ an effect lives for about $1/(1-\beta R)$ periods: about 420 in Deck 4.1's economy.
- A one-period condition (Methods 2 and 3) sees only the first bar.
- **DeepHAM sees $T_{\text{unroll}} = 150$ periods, through a sample; Feng, Han, Sargent & Zhu (2026), 100 or 30, exactly.**

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Source: Lab block B, the exact histogram derivative; the lab's annual economy. Feng, Han, Sargent & Zhu (September 2026), preliminary.

Bridge: Each Tool for Its Question

Sequence space: long memory, stable training, an exogenous domain.

Deep Equilibrium Nets: the whole histogram as state; first-order conditions.

DeepHAM: interpretable moments; long rollouts.

Feng, Han, Sargent & Zhu (2026): the law of motion in the graph; a government without commitment.

- For the competitive equilibrium of this economy, all three work, and the classical rule is already accurate.
- The difference opens when someone chooses a policy that moves the distribution, and must be consistent with later choices.
- **Deck 4.3: households that age, and aggregate risk over cohorts, where the third method earns its place.**

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Source: Feng, Han, Sargent & Zhu (September 2026), preliminary; the sources of this deck.

Session 4.3

Cohorts Under Aggregate Risk

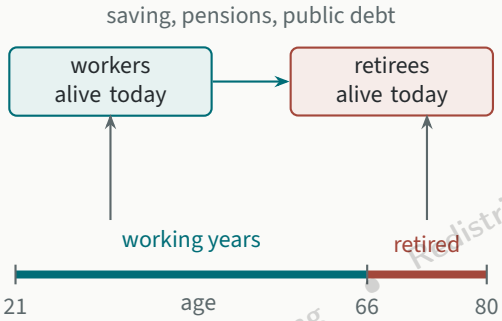
From Krusell–Smith to overlapping generations: a finite state, a network that clears a market, and why idiosyncratic risk sends us to histories



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From Krusell–Smith to Cohorts



- Deck 4.2: households live for ever; the state is $(a, \varepsilon; z, \Gamma)$ and Γ must be forecast.
- **Overlapping generations** (OLG): lives last J periods, and cohorts of every age $j = 1, \ldots, J$ coexist.
- Newborns hold nothing, the oldest leave, and Γ becomes a *stack*: one distribution Γ_j for each age.
- **Same question as Deck 4.2, what stands for Γ, with a new structure to exploit.**

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Source: Samuelson, *JPE* (1958); Diamond, *AER* (1965); Auerbach & Kotlikoff, *Dynamic Fiscal Policy* (1987).

Two Kinds of Risk, Four Economies

	no aggregate risk	**aggregate risk z_t**
idiosyncratic ε	**Part A**: the classical life cycle; Γ a stack of cohort histograms; paths after one shock	**Part C**: every cohort a histogram, J of them moving with z; the state explodes
no ε	one household per cohort, no risk: a deterministic life cycle	**Part B**: one household per cohort; the state is $1 + 2J = 15$ numbers; a network

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- Parts A and B each drop one kind of risk, and each can then carry Γ exactly.
- A path answers one unanticipated shock. With z_t drawn every period the policy must read (z, Γ): Krusell and Smith's problem, now with a stack.
- **Part C puts both risks back, and borrows Deck 4.2's third method to survive it.**

Source: Krusell & Smith, *JPE* (1998); Ríos-Rull, *RES* (1996); Azinovic-Yang & Žemličková (2025).

Finite Lives: The Last Period Is Known



each arrow is one endogenous-grid step, using next year's policy

- The lab's life-cycle economy: $J = 60$ cohorts, ages 21 to 80, idiosyncratic ε, a limit $a' \geq 0$, a pension b.
- The policy depends on age: $a' = g_j(a, \varepsilon)$. At the last age consumption is cash on hand.
- Given c_{j+1}, Deck 4.1's Euler step delivers c_j: 59 steps in all, from age 80 back to 21. Deck 4.1's infinitely lived household instead iterates to a fixed point: 356 sweeps from a cold start (the Session 4 lab, block A).
- **Backward induction replaces iteration to convergence: the household problem has no fixed point left.**

Source: Auerbach & Kotlikoff (1987); Carroll, *Economics Letters* (2006); the Session 4 lab, block C2. Annual: $\alpha = 0.33$, $\beta = 0.96$, $\delta = 0.08$, $\sigma = 2$, $\rho_\varepsilon = 0.90$, $a' \geq 0$.

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Cohorts Are the Distribution



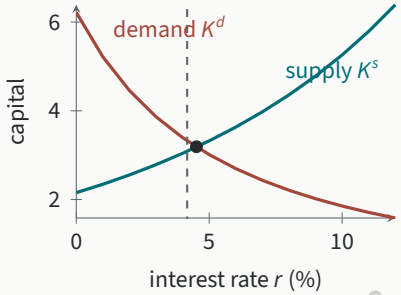
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- Γ_1 is known: newborns hold nothing, with ε drawn from its invariant distribution.
- $\Gamma_{j+1} = \Gamma_j \Phi_{g_j}$: Deck 4.1's lottery matrix, applied once for each age.
- Given prices, no iteration to an invariant distribution: 59 pushes complete the cross-section. The price is still unknown: next page.
- **The economy's distribution is the stack of the cohorts'. Part C will need exactly this stack, moving with z.**

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Source: Young, *JEDC* (2010); Ríos-Rull, *Review of Economic Studies* (1996).

The Stationary Equilibrium: One Price, Found by Bisection



The stationary equilibrium

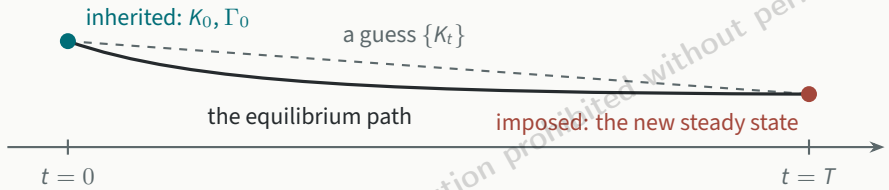
1. bracket $r \in [-7\%, 30\%]$; check the sign change
2. trial r: firm gives K/L, w; tax pays pension
3. households backward: 59 EGM steps
4. cohorts forward: 59 pushes give $K^s(r)$
5. compare with $K^d(r)$; halve the bracket
6. stop after 29 halvings; residual 6×10^{-9}

- $r^* = 4.52\%$ lies above $1/\beta - 1 = 4.17\%$ (dashed): with finite lives that bound fails; the bracket must be wide.
- **One price, one fixed point; inside it, one backward and one forward pass.** Code: the Session 4 lab, block C2.

Source: The Session 4 lab, block C2: 60 cohorts, a 40% pay-as-you-go pension.

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The Unknown Is a Path of Prices

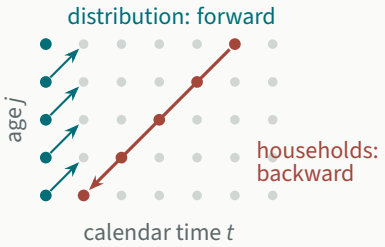


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- An unanticipated, permanent change at $t = 0$: an "MIT shock". Afterwards the path is known.
- Unknown: $\{K_t\}_{t=0}^{T}$. Prices $\{r_t, w_t\}$ follow from the firm's conditions.
- Boundary conditions at both ends: Γ_0 is inherited, and behaviour at T is the new steady state's.
- **Decisions look forward and distributions look back. No single pass can do both.**

Source: Auerbach & Kotlikoff (1987); Conesa & Krueger, *RED* (1999).

Update, Damp, Repeat



The transition loop

1. solve both steady states; choose the horizon T
2. guess $\{K_t\}$; prices follow from it
3. households backward along each cohort's diagonal
4. distribution forward from Γ_0: the supply K_t^s
5. stop when $\max_t |K_t^s - K_t|/K_t$ is small
6. else $K_t \leftarrow (1-d)K_t + dK_t^s$, K_0 fixed

- Damping $d = 0.5$: a higher guess lowers r_t and the saving meant to support it, so a full step can oscillate.
- the Session 4 lab, block C runs this loop for Deck 4.1's infinitely lived economy after a 20% tax on capital income: 91 passes to 1×10^{-7}, $T = 300$. **The outer loop is a fixed point in a sequence.**

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Source: Auerbach & Kotlikoff (1987); Conesa & Krueger (1999); the Session 4 lab, block C. Annual: $\alpha = 0.33$, $\beta = 0.96$, $\delta = 0.08$, $\sigma = 2$, $\rho_\varepsilon = 0.90$, $a' \geq 0$.

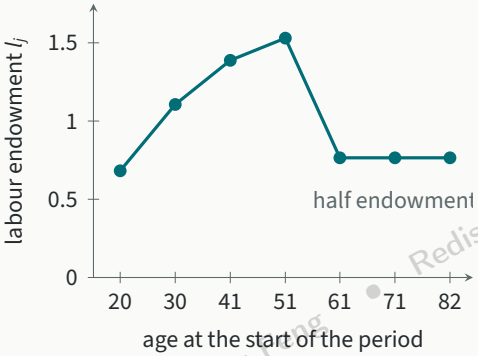
Measuring Who Gains

- **Consumption-equivalent variation** $g(a, \varepsilon)$: scale the old consumption plan by $(1+g)$ at every date and in every state, until its value matches the new one. $g > 0$ is a gain.
- With CRRA utility V scales by $(1+g)^{1-\sigma}$, so $g(a, \varepsilon) = \left[V_1(a, \varepsilon)/V_0(a, \varepsilon)\right]^{1/(1-\sigma)} - 1$.
- Everything turns on *which* V_1. Along the path: the value at $t = 0$ of living through the transition. Between steady states: the value of waking up in the new one.
- Average with the distribution of the people who are asked: Γ_0, the households alive at the reform.
- **One formula, two numbers. The difference is whether the path is counted.**

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Source: Lucas, *Models of Business Cycles* (1987); Domeij & Heathcote, *International Economic Review* (2004).

The Economy: Seven Cohorts, Two Assets

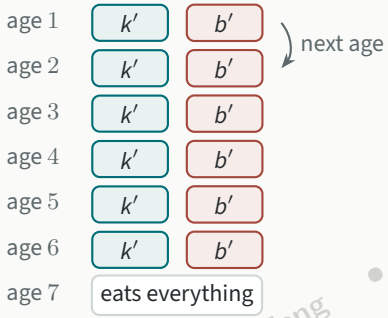


- $J = 7$ cohorts; one period is $72/J \approx 10.3$ years, ages 20 to 82. Endowments l_j (left), summing to J.
- CRRA $\gamma = 2$; $\beta = 0.96^{72/J} \approx 0.66$. Two assets: capital k and a one-period bond b.
- TFP: $\log z' = \rho \log z + \sigma\epsilon$, $\rho \approx 0.19$, $\sigma \approx 0.056$: annual 0.85 and 0.03, rescaled.
- Depreciation moves with TFP: $\delta_t = \min\{\delta \frac{2}{1+z_t}, 1\}$, $\delta \approx 0.66$ a period (10% a year). **A bad draw also destroys capital.**

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Source: Part B follows Azinovic-Yang & Žemlička, "Economics-Inspired Neural Networks with Stabilizing Homotopies", arXiv 2303.14802, and its OLG notebook (the calibration cells). In PyTorch, scaled for a CPU: the Session 4 lab, block G.

The Household: Capital, Bonds, Adjustment Costs



- Age j with holdings (k, b) chooses (k', b'):

$$c + p\,b' + k' + \psi_k (k' - k)^2 = l_j w + b + (1 - \delta + r)\,k.$$

- $k' \geq 0$; $b' \geq b_{\min} = -0.05$; the last cohort saves nothing. $\psi_k = 0.1$: changing one's *own* capital is costly.
- $J - 1 = 6$ ages choose: $2(J-1) = 12$ saving decisions.
- **No ε: everyone of an age is identical, so a cohort is one household.**

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Source: Notebook markdown "Households" and the budget in *E*; *psi_k*, *b_min_target*.

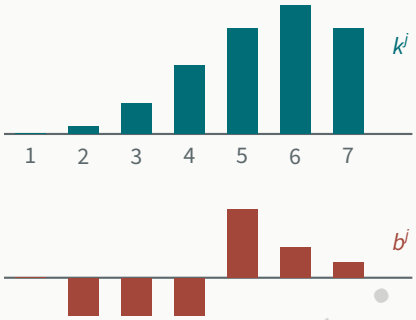
Firm and Markets

- Firm: $Y = z\,K^{\alpha}L^{1-\alpha}$, $\alpha = 1/3$; $r = \alpha Y/K$, $w = (1-\alpha)Y/L$.
- Aggregates: $K = \sum_j k^j$, $L = \sum_j l_j = J = 7$, fixed.
- Bonds in zero net supply: $\sum_j b'^j = 0$. The bond price p is the one price not given by a marginal product.
- Goods clear by Walras' law.
- The notebook's text swaps the formulas for r and w; the code, *get_firm_side*, is as written here.
- **Two assets, one of them priced by the market: a network must now return a price as well as policies.**

Source: Notebook *get_firm_side*; market-clearing markdown.

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The State Is Fifteen Numbers



age j; bars schematic

- The distribution is a vector, $\Gamma = (k^1, \ldots, k^J, b^1, \ldots, b^J) \in \mathbb{R}^{2J}$; the input $x = (z, \Gamma)$ has $d = 1 + 2J = 15$ numbers.
- Unknowns: $J - 1$ capital and $J - 1$ bond policies and the price, as functions of x: $n_{\text{out}} = 13$.
- Equilibrium: $2(J - 1) = 12$ KKT conditions (six saving ages, two assets) and bond clearing, at every x.
- **Unlike Deck 4.2: Γ is exact. No moment, no histogram, no history.**

Source: Notebook “State Variable and Equilibrium Functions”; the index constants; the code keeps $k^1 = b^1 = 0$ as inputs.

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Tomorrow's Distribution Is a Shift



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- Age j's choice today is age $j+1$'s holding tomorrow; a newborn enters with $(0, 0)$; age J leaves. So $\Gamma' = (0, k'^{1}, \ldots, k'^{J-1}, 0, b'^{1}, \ldots, b'^{J-1})$: $2J$ numbers again.
- Γ' is known once today's choices are made: the only randomness in $x' = (z', \Gamma')$ is z', one successor per quadrature node.
- **Unlike Deck 4.2's lottery: no mass to split between grid points. The policy's output *is* the next state.**

Source: Notebook *ForwardStep* and the "next period wake-up assets" in *E*.

Equilibrium Conditions as Fischer–Burmeister Residuals

- KKT conditions for age $j < J$, written as relative consumption errors, u'^{-1} the inverse marginal utility:

$$\varepsilon_k = \psi^{FB}\Big(\frac{u'^{-1}\big(\beta\,\mathbb{E}[u'(c')\,(1-\delta'+r'+2\psi_k(k''-k'))]\,/\,(1+2\psi_k(k'-k))\big)}{c} - 1,\ \frac{k'}{c}\Big),$$

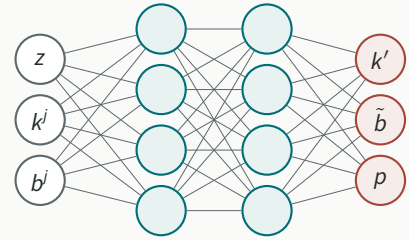
$$\varepsilon_b = \psi^{FB}\Big(\frac{u'^{-1}\big(\beta\,\mathbb{E}[u'(c')]\,/\,p\big)}{c} - 1,\ \frac{b'-b_{\min}}{c}\Big), \qquad \psi^{FB}(a,b) = a + b - \sqrt{a^2+b^2}.$$

- $\psi^{FB} = 0$ exactly when $a \geq 0$, $b \geq 0$, $ab = 0$: the Euler equation holds, or the limit binds.
- $\mathbb{E}$: $n_{GH} = 8$ Gauss–Hermite nodes over z'. c' and k'' from the same network at x'.
- **As Deck 4.2's Deep Equilibrium Nets, one period ahead; now two assets and $2(J-1) = 12$ residuals per state.**

Source: Notebook *E* (capital and bond terms), *fischer_burmeister*, *num_GH_tfp*; Maliar, Maliar & Winant, *JME* (2021).

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The Network: Fifteen Numbers In, Thirteen Out

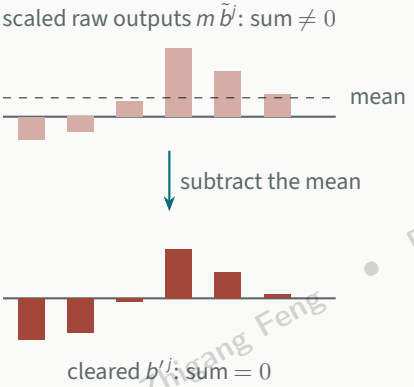


- One network for all $n_{\text{out}} = 13$ functions: two shared selu layers of $n_{\text{hid}} = 10J = 70$ units, so ages share what they learn.
- Capital: $k' = 0.1\,\text{softplus}(o_k) \geq 0$, so no short sales by construction.
- Price: $p = \text{softplus}(o_p) > 0$.
- Bonds: the raw $\tilde{b}$ goes through a clearing layer (next page).
- **Consumption is not an output: it is backed out of the budget.**

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Source: Notebook *base_net_construct*, *S_fun*; hidden width $N = 10\times N_cohort$.

A Layer That Clears the Bond Market



- From the $J-1=6$ raw outputs $\tilde{b}^j$:

$$b'^j = m\,\tilde{b}^j - \frac{1}{J-1}\sum_i m\,\tilde{b}^i.$$

- Each age keeps its scaled output minus the six-age average. Summed over j, the terms cancel: $\sum_j b'^j = m\sum_j \tilde{b}^j - m\sum_i \tilde{b}^i = 0$.
- m: set by hand, not learned (next page).
- **Unlike Deck 4.2: the market clears by construction, not by a loss term.**

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Source: Notebook *S_fun*: the mean excess over the net supply (zero here) is subtracted; the paper (market-clearing layers).

The Mask and the Price: One Set by Hand, One Pinned by the Loss

stages	m	**bond positions** $b'^j = m\,(\tilde{b}^j - \bar{\tilde{b}})$, $\bar{\tilde{b}}$ the six-age average
1–2	0	all zero, whatever the network outputs: no bonds yet
3 to the end	0.1	the raw outputs shrunk tenfold: bonds enter small

- **Mask** m: not an output, not trained. One of five numbers set by hand at each stage of training (the homotopy, later): bond supply 0, the limit $b_{\min}$, m, and two loss weights.
- Why shrink: freshly switched on, the bond outputs are untrained; scaled by 0.1 they barely move the economy just solved. Afterwards 0.1 is a fixed scale.
- **Who sets** p**?** Not clearing, which now holds at any p. The $J - 1 = 6$ bond Euler residuals do: six conditions for six unknowns, p and the positions, which have five free directions once they sum to zero.
- **A price is an output pinned down by the loss. With no bonds traded, nothing pins it.**

Source: Notebook *S_fun*, the *Homotopy* array, the stage cells; a comment says 0.01, the code sets 0.1.

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One Call at One State: Every Cohort's Choice, Then Every Budget

age j	holds today	chosen by the network	consumes, from its budget
1	$(0, 0)$: newborn	k'^1, b'^1	c^1
$2, \ldots, 6$	(k^j, b^j)	k'^j, b'^j	c^j
7	(k^7, b^7)	nothing: $k'^7 = b'^7 = 0$	c^7: all that is left
all ages	z; then K, r, w	one price p	

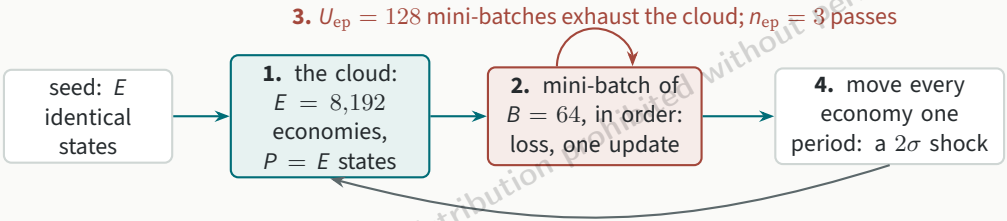
$$c^j = l_j w + b^j + (1 - \delta + r)\, k^j - p\, b'^j - k'^j - \psi_k \left(k'^j - k^j\right)^2, \qquad j = 1, \ldots, J.$$

- One call at one state $x = (z, \Gamma)$ returns all $n_{\text{out}} = 13$ numbers: every saving age's k'^j, b'^j and p. A cohort is one household, so this is every household's choice.
- No maximisation, no EGM: k'^j is the network's guess; the Euler residuals judge it.
- **Unlike Deck 4.2's Method 2, whose network maps one household's $(\varepsilon, a; z, \Gamma)$ to its (a', λ): here one call is the whole economy.**

Source: Notebook *S_fun* (the 13 outputs) and the budget in *E*; as in the code, $k^1 = b^1 = 0$ and the last cohort saves nothing.

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The Training Loop in One Picture

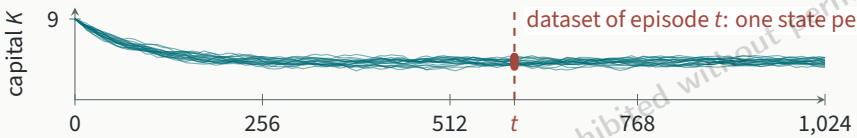


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- **1. Samples**: $E = 8{,}192$ economies side by side; their current states are the dataset.
- **2. Mini-batch**: the next $B = 64$ states in order, the loss at them, one gradient step.
- **3. Epoch**: $U_{\text{ep}} = E/B = 128$ mini-batches use the cloud once; $n_{\text{ep}} = 3$ passes.
- **4. Episode**: every economy moves one period; stage 1 alone runs $n_1 = 1{,}024$ episodes.

Source: Notebook *CloudMethod*, *Episode*, *TrainingLoop*; *n_cloud*, *n_minib*, *epochs_epis*. Symbols: the symbol page.

1. Samples: 8,192 Economies From One Seed, Never Restarted



- Each line is one economy's capital; 20 of the $E = 8{,}192$ are drawn. All start identical: $\log z = 0$, $k^2, \ldots, k^7 = 1.5$ ($K = 9$), no bonds; they differ only by their shocks.
- Dataset of episode t: the slice, $P = E$ states, one per economy, each $x = (z, \Gamma)$.
- Train, then move: $U_{\text{epis}} = 384$ updates on the slice, then every economy moves one period with the updated network. The seed is used once; no run restarts.
- **While the network trains on slice t, only that slice exists: earlier ones are discarded, later ones not yet simulated.**

Source: Notebook settings ($n_cloud = 2^{13}$), the seed (*representative_input*), *Episode*, *CloudMethod*. Paths schematic.

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2–3. Mini-Batches in Order, the Cloud Exhausted, Three Passes

One episode's cloud, then $U_{\text{epis}} = 384$ updates

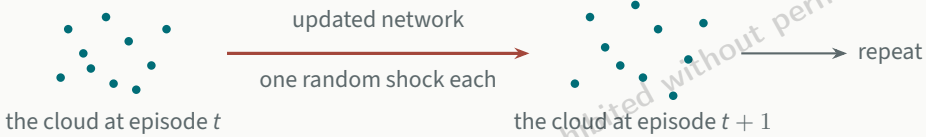
1. the cloud: $P = E = 8{,}192$ states $x = (z, \Gamma)$, one per economy, in the economies' order
2. take the next $B = 64$ states in order: economies 1–64, then 65–128, and so on
3. one Adam update of θ on the loss of the next pages, rate 10^{-5}
4. repeat 2–3; after $U_{\text{ep}} = E/B = 128$ updates every state is used once: *exhausted*
5. twice more, the same mini-batches in the same order: $n_{\text{ep}} = 3$ passes, $U_{\text{epis}} = n_{\text{ep}} U_{\text{ep}} = 384$
6. then the cloud moves one period (next page); the old states are discarded

- **Epoch**: one pass over the cloud, U_{ep} updates. Each state enters the loss $n_{\text{ep}} = 3$ times, never shuffled; Deck 4.2's Method 2 used each point once, DeepHAM's critic reused its rows 201 times.
- **One Adam update of θ per mini-batch:** $U_{\text{epis}} = n_{\text{ep}} E/B = 384$ **an episode;** $n_1 U_{\text{epis}} = 1{,}024 \times 384 = 393{,}216$ **in stage 1.**

Source: Notebook *TrainingLoop* ($n_minib = 2^6$, $epochs_epis = 3$; the cloud repeated and reshaped in order), *Update*.

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4. Episodes: the Cloud Moves One Period



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- After the $U_{\text{epis}} = 384$ updates, each economy moves one period: its policy from the updated network, its own shock drawn at $m_\sigma = 2$ times σ, truncated at ± 3 s.d. (the loss uses σ, so the data reach past the economy's usual range).
- The moved states replace the old ones and are the next episode's dataset.
- Stage 1, the capital-only economy, runs $n_1 = 1{,}024$ episodes; the later stages continue from the same E economies.
- **This is the only simulation step. The look-ahead inside the loss (next page) computes an expectation and moves nothing.**

Source: Notebook *Episode* (*truncated_normal*, *shock_mult* $= 2$), *ForwardStep*, *CloudMethod*; Bretscher et al. (2022).

5. Inside One Update: One Mini-Batch, Line by Line

One update on a mini-batch of $B = 64$ states

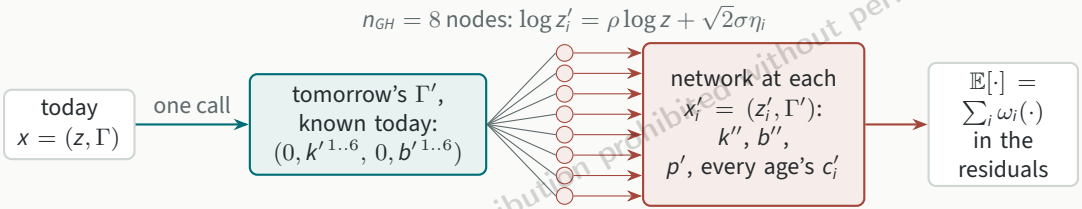
1 take the next $B = 64$ states $x = (z, \Gamma)$ of the cloud, in order
2 for each state, one call gives every saving age's k'^j, b'^j and p; each c^j from its budget
3 the same economy tomorrow at $n_{GH} = 8$ TFP nodes: $x'_i = (\rho \log z + \sqrt{2}\sigma\eta_i,\ \Gamma'), \Gamma'$ shared
4 call the network at each x'_i: k'', b'', p'; then every age's c'_i from the budget at $t + 1$
5 $\mathbb{E}[\cdot] = \sum_i \omega_i(\cdot)$; the residuals ε_k, ε_b for ages $1, \ldots, J - 1$
6 mean loss over the B; gradient through all $1 + n_{GH} = 9$ calls, k', b' inside x'_i too; Adam, 10^{-5}

- Steps 3–5 are the look-ahead (next two pages): eight successors of one economy, not households, not simulated. Per state $1 + n_{GH} = 9$ calls; per update $n_{\text{call}} = (1 + n_{GH})B = 576$.
- **Unlike Deck 4.2's Method 2, whose successors are stored data: here they are rebuilt inside the loss, so the gradient sees the move to x'.**

Source: Notebook *E*, *vE*, *Loss*, *GradAndLoss*, *Update*; *num_GH_tfp* $= 8$.

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5. The Look-Ahead: One Economy, Eight Successors

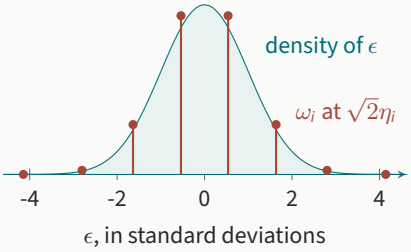


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- A successor is tomorrow's state of this economy at one TFP node: $x_i' = (\log z_i', \Gamma')$, $i = 1, \ldots, n_{GH}$; only z' differs across the eight.
- $\Gamma' = (0, k'^1, \ldots, k'^6, 0, b'^1, \ldots, b'^6)$ in all eight: newborns hold nothing, age j's choice is age $j+1$'s holding, age 7 has left.
- **Not households, unlike Deck 4.2's $n_{\text{pt}} = 8$ evaluation points, and not simulated: they serve the expectation (next page), then are discarded.**

Source: Notebook *E*: *x_next* stacks the eight successors in an 8×15 array whose rows differ only in the first entry.

5. The Look-Ahead: Eight Gauss–Hermite Nodes



$$\mathbb{E}[g(\epsilon)] \approx \sum_{i=1}^{n_{GH}} \omega_i \, g\left(\sqrt{2}\,\eta_i\right)$$

- The shock: $\log z' = \rho \log z + \sigma\epsilon$, $\epsilon \sim N(0,1)$.
- η_i: roots of the Hermite polynomial H_8; ω_i: its weights over $\sqrt{\pi}$, summing to 1.
- $\epsilon = \sqrt{2}\eta$ turns the normal density into the rule's weight $e^{-\eta^2}$: hence the $\sqrt{2}$.
- Exact for polynomials of degree $\leq 2n_{GH} - 1 = 15$.
- In the residuals: $\mathbb{E}[u'(c'^{j+1})] = \sum_i \omega_i \, u'(c_i'^{j+1})$.
- **Fixed nodes: no sampling noise.**

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Source: Notebook *hermgauss* (NumPy) for *eta_tfp*, *GH_weights_tfp*; the plotted nodes and weights are computed.

5. Inside One Update: the Loss

- A mini-batch $\mathcal{B}$ is $B = 64$ states x (a setting, 2^6); residuals are averaged over the B states and the $J - 1$ ages:

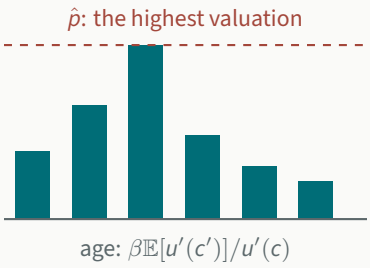
$$\mathcal{L}(\theta; \mathcal{B}) = \tfrac{1}{2}\,\overline{\varepsilon_k^2} + \tfrac{1}{2}\,w_b\,\overline{\varepsilon_b^2} + w_{\text{pre}}\,\overline{\varepsilon_{\text{pre}}^2} + 100\,\overline{\varepsilon_c^2}.$$

- $\varepsilon_c = \max(c, 10^{-4}) - c$: a network that implies negative consumption is pushed back hard.
- $\varepsilon_{\text{pre}} = p - \text{sg}(\hat{p})$: a target price, with its gradient stopped.
- w_b, w_{pre} and the mask m are switched by the stage. One Adam step per mini-batch: $U = 3{,}970{,}560$ in all.
- **Unlike Deck 4.2's methods, the loss weights change during training: the loss is a path of problems.**

Source: Notebook *Loss*, *GradAndLoss*, *E* (*eps_cnn*, *eps_bpr*).

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Why a Homotopy: a Bond Nobody Can Trade Has No Price

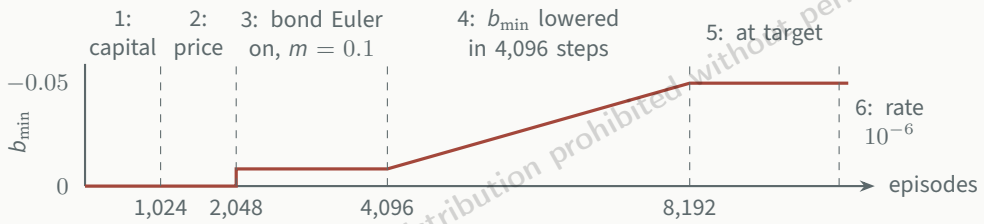


- At $b_{\min} = 0$ and zero net supply no one can hold a bond, so any high enough price clears the market: p is not pinned down.
- Its natural limit: the **zero-liquidity price** $\hat{p} = \max_j \beta\,\mathbb{E}[u'(c'^{j+1})]/u'(c^j)$, what the most eager saver would pay for the first bond.
- Then relax $b_{\min}$ a little at a time; each problem starts from the last one's solution.
- **Solving the full model from a random start is where state-space training tends to fail.**

Source: Notebook "Pretrain bond price", *p_b_implied*; the paper.

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The Homotopy in Six Stages



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- 1: capital only, $w_b = w_{\text{pre}} = m = 0$. 2: $w_{\text{pre}} = 1$, learn $\hat{p}$. 3: $w_b = 1$, $w_{\text{pre}} = 0$, $m = 0.1$, $b_{\min} = -0.05/(J-1)$.
- 4: $b_{\min}$ lowered to -0.05, one episode a step. 5: at the target. 6: the rate cut to a tenth.
- **Each stage inherits network, cloud and optimiser: the same $E = 8{,}192$ economies run through all six. Episodes and updates: next page.**

Source: Notebook, “Initialize Capital-Only Model” to fine-tuning; a comment says mask 0.01, the code sets 0.1.

The Six Stages in Episodes and Updates

stage s	what is switched	n_s	$U_{\text{epis}}\, n_s$	rate
1 capital only	$w_b = w_{\text{pre}} = m = 0$; from the seed	1,024	393,216	10^{-5}
2 learn the price	$w_{\text{pre}} = 1$: p fitted to $\hat{p}$	1,024	393,216	10^{-5}
3 bonds on	$w_b = 1, w_{\text{pre}} = 0, m = 0.1, b_{\min} = -0.05/6$	2,048	786,432	10^{-5}
4 lower the limit	$b_{\min}$ down to -0.05, one episode a step	4,096	1,572,864	10^{-5}
5 at the target	nothing: train on at $b_{\min} = -0.05$	2,024	777,216	10^{-5}
6 fine-tune	the rate only	124	47,616	10^{-6}
all	$n_{\text{epis}} = \sum_s n_s$; $U = n_{\text{epis}} U_{\text{epis}}$	10,340	3,970,560	

- $U_{\text{epis}} = 384$ updates an episode throughout, so stage s makes $U_{\text{epis}} n_s$ updates: $U = n_{\text{epis}} U_{\text{epis}} = 10{,}340 \times 384 = 3{,}970{,}560$ in all.
- **The notebook times each stage in minutes on one T4 GPU, in comments; nothing else about the run is stored.**

Source: Notebook *cloud_steps_cap*, *cloud_steps_first*, *homotopy_steps*, the final and fine-tuning calls; *Gamma*, *Gamma_fine*.

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Three Clocks, and the Algorithm

clock	every	what is simulated, from where, and what it feeds
network	each mini-batch of	nothing simulated: $n_{GH} = 8$ successors inside the loss,
update	$B = 64$, in order	discarded; one Adam step on θ
episode	$U_{\text{epis}} = 384$ updates	one period of all E from their own states, network frozen, 2σ shocks; the moved cloud replaces the old
stage	after its n_s episodes	nothing: $b_{\min}$, m, w_b, w_{pre} or the rate switch; network, cloud and optimiser carry over

The algorithm: $U = 3{,}970{,}560$ updates

1 seed $E = 8{,}192$ identical states and a random network
2 for each stage s, set $(b_{\min}, m, w_b, w_{\text{pre}}, \text{rate})$; for each of its n_s episodes:
3 $\quad U_{\text{epis}} = 384$ updates: $n_{\text{ep}} = 3$ passes of $U_{\text{ep}} = 128$ mini-batches of $B = 64$ states
4 $\quad$ move the cloud one period with fresh shocks at 2σ

Source: Notebook *CloudMethod*, *Episode*, *TrainingLoop*; the stage cells.

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The Symbols

symbol	value	what it counts, and the formula
J	7	cohorts (*N_cohort*); one period is $72/J \approx 10.3$ years
$\beta, \delta, \rho, \sigma$	0.66, 0.66, 0.19, 0.056	$0.96^{72/J}$; $1-0.9^{72/J}$; $0.85^{72/J}$; $0.03\sqrt{\sum_{i<10} 0.85^{2i}}$
d, n_{out}	15, 13	inputs (z, Γ), $1+2J$; outputs, $2(J-1)+1$
n_{hid}, n_{GH}	70, 8	selu units per hidden layer, $10J$; quadrature nodes for z'
E, P	8,192, 8,192	economies side by side, 2^{13}; the dataset, $P = E$
B, U_{ep}	64, 128	mini-batch, 2^6, in order; updates that use the cloud once, E/B
n_{ep}, U_{epis}	3, 384	passes over the cloud, same order; updates an episode, $n_{\text{ep}} U_{\text{ep}}$
n_{call}, m_σ	576, 2	network calls per update, $(1+n_{GH})B$; shock multiple
n_{epis}	10,340	episodes in all; stage s runs n_s: $\sum_s n_s = 1{,}024 + 1{,}024 + 2{,}048 + 4{,}096 + 2{,}024 + 124$
U	3,970,560	updates in all, $n_{\text{epis}} U_{\text{epis}}$
m, $b_{\min}$	0.1, -0.05	mask on bonds; the limit, from $-0.05/(J-1)$ in 4,096 steps
rate	$10^{-5}, 10^{-6}$	Adam; in the last 124 episodes

Source: Notebook settings *N_cohort*, *n_cloud*, *n_minib*, *epochs_epis*, *num_GH_tfp*, *shock_mult*, the stage cells, *Gamma*.

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Four Methods, One Table

	DeepHAM	Deep Eq. Nets, KS	Sequence space	Deep Eq. Nets, OLG (here)
what stands for Γ	learned moments of $N = 50$	two $N_b = 100$-point histograms	the last $T_{\text{hist}} = 50$ shocks	Γ **itself:** $2J = 14$ **numbers**
what is minimised	minus lifetime utility; a value regression	Euler and KKT residuals	an FB Euler residual	FB residuals; bond clearing built in
where states come from	$E = 384$ panels, refreshed	$E = 16$ tracks, carried forward	a pool of $P_{\max} = 128$	a cloud of $E = 8{,}192$, moved one period
what the gradient passes through	$T_{\text{unroll}} = 150$ periods, simulated	one period, at stored states	one period; prices fixed	one period, through the computed x'

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- **Row 1 is why the OLG is easy here: with no ε, nothing has to stand for Γ.**

Source: Deck 4.2, *Four Questions Every Deep Method Answers* and *Three Pipelines, the Same Letters*; the homotopy notebook.

What Changed From Deck 4.2

- **An exact state.** Cohorts without ε are points, so the distribution is a vector of $2J = 14$ numbers, and its law of motion is a shift.
- **A price as an output.** Two assets need a bond price; the network returns it, and a layer clears the market exactly.
- **A path of problems.** Capital first, then a price, then bonds that trade a little more each step.
- **Wide and shallow simulation.** $E = 8{,}192$ economies moved one period at a time, at twice the shock's s.d.
- **Successors differentiated.** The loss builds x' from the policy; the gradient sees the move.
- **All five rest on the first. Take it away and the rest has nothing to stand on.**

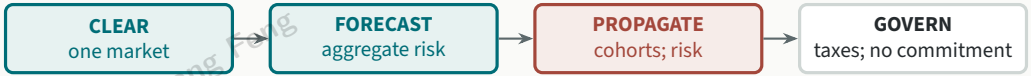
Source: The homotopy notebook; Deck 4.2.

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Session 4.3

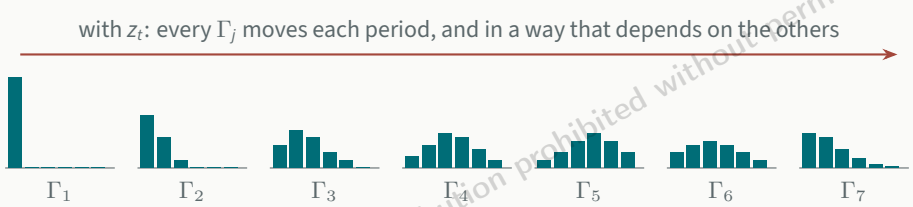
Put Idiosyncratic Risk Back

After Azinovic-Yang & Žemlička, *Deep Learning in the Sequence Space*: every cohort becomes a distribution, and the state explodes



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Each Cohort Becomes a Histogram

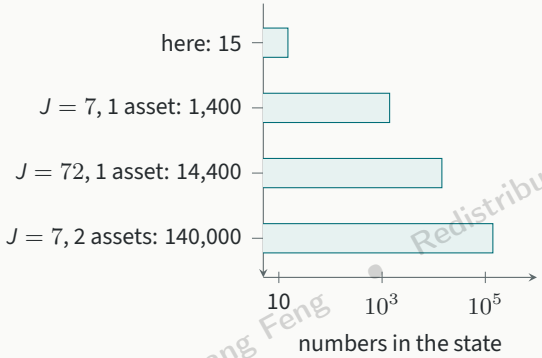


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- With ε, two households of the same age differ. A cohort is a distribution $\Gamma_j(\varepsilon, k)$ again, as in Part A.
- The aggregate state: $(z, \Gamma_1, \ldots, \Gamma_J)$. Part A's stack, but no longer stationary: it moves with z.
- **Part A had the stack without z; Part B had z without the stack. Now both.**

Source: Part C follows Azinovic-Yang & Žemlička, "Deep Learning in the Sequence Space", working paper, arXiv 2509.13623 (2025), §6, with their slides (2026); here p. 113. In PyTorch, scaled for a CPU: the Session 4 lab, block H.

Count the Inputs



- Two income states, $N_b = 100$ asset points per cohort: $J \times 2 \times N_b = 7 \times 2 \times 100 = 1{,}400$.
- With the published OLG's $J = 72$ cohorts: $72 \times 2 \times 100 = 14{,}400$.
- Two assets on a 100×100 grid: $7 \times 2 \times 10^4 = 140{,}000$.
- **A state-space network reads all of it, and its training states move whenever the policy does.**

Source: Arithmetic. Cohort count: their slides, p. 13.

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Why Deck 4.2's First Two Answers Strain

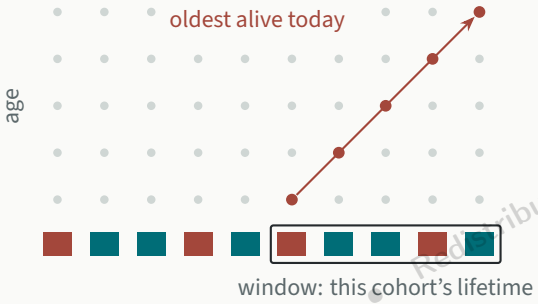
	Krusell–Smith, one distribution	**OLG, J distributions**
classical rule	forecast K' from K	forecast every cohort's moments, or prices from many of them
DeepHAM	one panel of $N = 50$, one learned moment	a panel for each age; moments per age
Deep Eq. Nets	$2N_b = 200$ masses as input	$J \times 2N_b$ masses, drifting with the policy
sequence space	the last $T_{\text{hist}} = 50$ shocks	**the same window of shocks**

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- The first three grow with the number of ages J; the last does not, because its input never contained Γ.
- Deck 4.2, *What the Exogenous Domain Buys*: large inputs that shift with the policy feed the unstable loop.
- **Only the history keeps its size when cohorts are added.**

Source: Deck 4.2; their slides, pp. 11, 23–24.

The Way Out: the History Already Holds Every Cohort

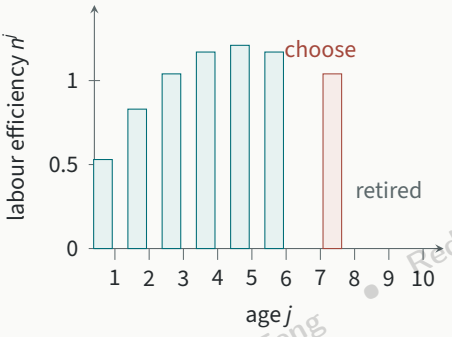


- Everyone alive was born within the last J periods: Γ_j is built by the shocks of its lifetime, from its starting wealth.
- Older shocks act only through prices and inheritances (bequests, in the model ahead): Deck 4.2's **memory loss**.
- So the stack is, approximately, a function of recent shocks: a window at least a lifetime long, $T_{\text{hist}} \geq J$.
- **Published sweep, 1 to 256 lags: accuracy improves with the window.**

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Source: Intuition ours; evidence: their slides, pp. 7–8, 20–21, 91 (the paper's 72-cohort OLG).

The Economy: Ten Cohorts, Income Risk, a Retirement Choice

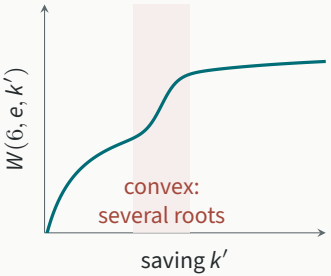


- $J = 10$ cohorts of six years: ages 1–6 work, age 7 chooses to work or retire (logit, scale 0.08), 8–10 are retired; bequests become newborns' assets.
- Income $e \in \{0.5, 1, 1.5\}$, $N_e = 3$, Markov; one asset, capital, $k \geq 0$ on $N_k = 300$ points.
- Aggregate risk: $n_z = 3$ regimes; the third is a disaster, depreciation 0.397 instead of 0.265.
- State (z, Γ), $\Gamma(j, e, k)$: $JN_eN_k = 9{,}000$ masses. Input: the last $T_{\text{hist}} = 100$ regimes.
- **The network never reads Γ.**

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Source: The paper, §6.1–6.2; their slides, pp. 105–113. The paper numbers ages 0–9; its implementation text says 200 asset points, its parameter table 300.

Why the Network Outputs Continuation Values

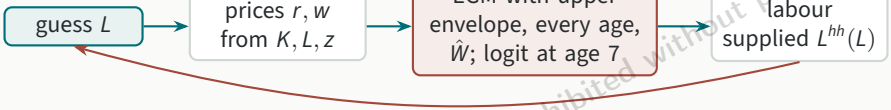


- Age 6 looks ahead to retirement, so its value of saving is convex in places (left): several Euler roots; a residual can pick the wrong one.
- So the network outputs **continuation values** $\hat{W}(j, e, k') \approx \mathbb{E}[V(j{+}1, e', k')]$ on the grid, ages $1, \ldots, J-1$; EGM with an **upper envelope** keeps the best root.
- Increasing by construction: a first value plus softplus increments. One call gives all $(J-1)N_eN_k = 8{,}100$ values: operator learning.
- **Unlike Deck 4.2's tutorial: no consumption output, no residual.**

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Source: The paper, §6.3; their slides, pp. 110, 114–120; Fella (2014); Druedahl & Jørgensen (2017).

Within the Period: Policies, and a Labour Market That Must Clear



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- K is fixed by Γ; L is not: whether age 7 works depends on the wage, and the wage on L.
- $n_{\text{Newton}} = 5$ steps, each an EGM solve of all J cohorts, ED' by automatic differentiation; start $L^{\text{work}} + \hat{\lambda}\, L^{7,\max}$, $\hat{\lambda}$ the network's guess of age 7's share.
- Result: prices, every age's saving rule $g^j(e,k)$, the work probabilities $p^{\text{work}}(e,k)$, and the realised share λ.
- **Unlike Part B, where a layer cleared the bond market exactly: here a price is found by iteration inside every evaluation.**

Source: The paper, §6.3 (Step 0 of the forward simulation) and its hyperparameters.

Moving the Distribution One Period

- Young's lottery with income transitions, for a saving rule g and masses $m(i,j)$ at (e_i, k_j):

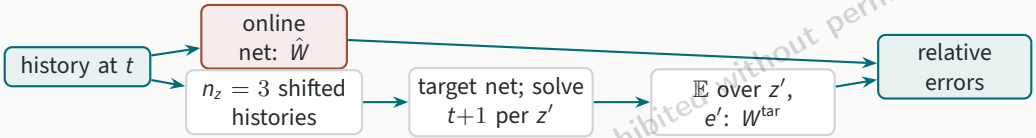
$$[\mathcal{T}(g,m)](i',j') = \sum_{i,j} m(i,j)\,\Pi^{e}_{ii'}\big(\omega^{-}\,\mathbf{1}\{j'=j^{-}\} + \omega^{+}\,\mathbf{1}\{j'=j^{+}\}\big),$$

  $j^{\pm}$ the grid points around $g(e_i,k_j)$, $\omega^{\pm}$ the lottery weights of Deck 4.2.
- Aging: $\Gamma'_{j+1} = \mathcal{T}(g^{j},\Gamma_j)$. Age 7 splits: $\mathcal{T}(g^{\text{work}}, p^{\text{work}} \odot \Gamma_7) + \mathcal{T}(g^{\text{ret}}, (1-p^{\text{work}}) \odot \Gamma_7)$.
- Newborns: $\Gamma'_1 = \mathcal{T}(g^{J},\Gamma_J)$, the bequests.
- Γ' does not depend on z': $n_z = 3$ candidate successors (z',Γ'), each with its history shifted by z'.
- **The same lottery as Deck 4.2's Methods 2 and 3, applied age by age.**

Source: The paper, §6.3 (Steps 1–3 of the forward simulation); Young, *JEDC* (2010).

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The Loss: Targets From a Slower Copy of the Network

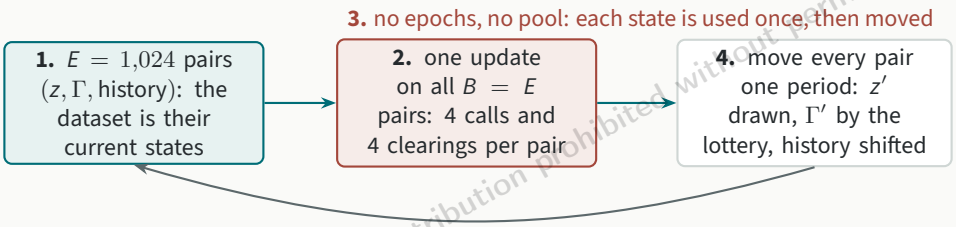


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- Target: $W^{\text{tar}}(j, e, k') = \sum_{z'} \Pi^{z}_{zz'} \sum_{e'} \Pi^{e}_{ee'} V_{t+1}(j{+}1, e', k' \mid z'; \bar{\rho})$, $t{+}1$ solved with the target net $\bar{\rho}$.
- Loss over the $B = 1{,}024$ pairs: mean squared relative errors $(\text{target} - \text{prediction})/(\varepsilon_{\text{loss}} + |\text{target}|)$ on W's first value, its increments and λ; $\varepsilon_{\text{loss}}$: 1 to 0.01.
- Targets fixed; one Adam step on ρ at 10^{-6}; then $\bar{\rho} \leftarrow \tau\rho + (1-\tau)\bar{\rho}$, $\tau = 10^{-4}$: the target lags about $1/\tau = 10{,}000$ updates (Deck 4.1: 0.02, 50).
- **A supervised target from a slower copy, as Deck 4.1's critic; not a residual.**

Source: The paper, §6.3 (loss function) and §6.4 (Stage 3, hyperparameters).

The Training Loop in One Picture



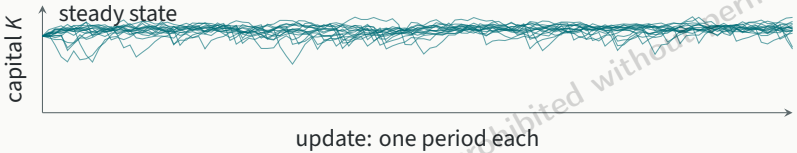
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- **1. Samples**: $E = 1{,}024$ pairs side by side; their current states are the dataset, $P = E$.
- **2. Update**: one Adam step on all $B = E = 1{,}024$ pairs: the mini-batch is the dataset.
- **3–4.** Nothing is reused: after the step every pair moves one period; the update is the period.
- **The look-ahead (n_z target calls, n_z clearings at $t + 1$) moves nothing.**

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Source: The paper, §6.4 (the outline and step 7). Symbols: this part's symbol page.

1. Samples: 1,024 Tracks, Moved One Period per Update



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- Start: Stage 1's steady state for all $E = 1{,}024$ pairs (the paper gives no initial histories); Stage 2 spreads them, continuation values held at the steady state's. 16 of E drawn.
- A pair's run never restarts: each update moves it one period (z' drawn, Γ' by the lottery, the history shifted): no pool, no branching, no episode. Dips: disasters.
- Coverage: histories drawn from Π^z meet the $3^{T_{\text{hist}}}$ windows at their true frequencies; a life is $J = 10$ periods, the window $T_{\text{hist}} = 100$; Γ comes along with its history.
- **Coverage is checked on simulated paths, not guaranteed.**

Source: The paper, §6.3 (Step 3) and §6.4 (Stages 1–2, step 7). Paths schematic.

2. Inside One Update, Line by Line

One Adam step on all $B = E = 1{,}024$ pairs $(z, \Gamma,$ history$)$

1. online network at each history: $\hat{W}$ and $\hat{\lambda}$ (one call)
2. target network at the $n_z = 3$ shifted histories (n_z calls)
3. clear labour at t: $n_{\text{Newton}} = 5$ steps, each an EGM solve of all J cohorts; prices, policies, λ
4. move Γ one period: n_z candidate successors
5. solve $t{+}1$ for each z' with the target network (n_z more clearings); the expectation: W^{tar}
6. loss; one Adam step at 10^{-6}; nudge the target network by τ
7. draw z' from Π^z; that successor is the pair's next state

- Per pair: $1 + n_z = 4$ network calls and $1 + n_z = 4$ clearings of $n_{\text{Newton}} = 5$ EGM solves each; times B: 4,096 calls and $(1 + n_z)\, n_{\text{Newton}} B = 20{,}480$ EGM solves an update.
- Three stages: (1) a steady state by EGM, which seeds the pairs; (2) aggregate risk on, continuation values held at the steady state, to spread them; (3) the loop above.

Source: The paper, §6.4 (the seven steps, Stages 1–3, hyperparameters).

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One Clock, Three Stages, and the Algorithm

clock	every	what is simulated, from where, and what it feeds
network update	every period, on all $B = E = 1{,}024$ pairs	per pair: 4 network calls, 4 clearings of $n_{\text{Newton}} = 5$ EGM solves; one Adam step on ρ, the target nudged by τ
stage	three in all	none: 1 a steady state seeds the pairs; 2 aggregate risk on, continuation values held at the steady state's; 3 the loop

The algorithm: the paper gives no count of updates

1. Stage 1: a deterministic steady state by EGM; $E = 1{,}024$ pairs seeded from it
2. Stage 2: aggregate risk on, continuation values held at the steady state's: the pairs spread
3. Stage 3, repeat: one update on all $B = E$ pairs (previous page); move each pair one period

- **Unlike Part B's episode of $U_{\text{epis}} = 384$ updates on one cloud and Deck 4.2's pool of $P_{\max} = 128$: here the update is the period, and nothing is reused.**

Source: The paper, §6.4 (Stages 1–3, step 7).

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The Symbols

symbol	value	what it counts, and the formula
J, N_e, N_k, n_z	10, 3, 300, 3	cohorts of six years; income states; asset points; regimes
JN_eN_k	9,000	masses in $\Gamma(j, e, k)$; 6,000 with 200 points
T_{hist}	100	regimes in the window (the paper's T); at least a lifetime, J
n_{out}	8,101	continuation values for ages $1, \ldots, J-1$, $(J-1)N_eN_k$, plus $\hat{\lambda}$
n_{hid}	$3 \times 1{,}024$	hidden layers (GELU)
E, B	1,024, 1,024	pairs $(z, \Gamma,$ history$)$ side by side; the mini-batch is all of them
n_{Newton}	5	Newton steps to clear labour; each an EGM solve
n_{call}, n_{clear}	4, 4	per pair and update: network calls, $1 + n_z$; clearings, $1 + n_z$
n_{EGM}	20	EGM solves per pair and update, $(1 + n_z)\, n_{\text{Newton}}$; $\times B$: 20,480
rate, τ, $1/\tau$	10^{-6}, 10^{-4}, 10,000	Adam; the target's nudge; updates the target lags, about
$\varepsilon_{\text{loss}}$	1, 0.1, 0.01	the stabiliser in the relative errors, lowered twice
stages, U	3, not given	steady state, pretraining, the loop; updates in all: not stated

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- The paper's H, h, μ and T are J, j, Γ and T_{hist} here.

Source: The paper, §6.2 (parameters), §6.3 (implementation), §6.4 (hyperparameters).

Two More Pipelines, the Same Letters

	Deep Eq. Nets, KS	Deep Eq. Nets, OLG (B)	Sequence space, KS	Sequence space, OLG (C)
side by side	$E = 16$ tracks	$E = 8{,}192$ economies	$P_{\max} = 128$ pool	$E = 1{,}024$ pairs
a run starts from	its own end state	one seed, no restart	a parent from the pool	the steady state, no restart
periods per run	$T_{\text{epis}} = 128$	one	$T_{\text{gen}} = 100$	one
B of *P*	128 of 16,384	64 of *E*, in order	16 of ≤ 128	all *E*
per dataset	$U_{\text{epis}} = 128$, exhausted	$U_{\text{epis}} = 384$: 3 passes	$U_{\text{round}} = 10$, never exhausted	1, used once
datasets	65,000 episodes	10,340 episodes	10,000 rounds	every update
U in all	8,320,000	3,970,560	100,000	not given
look-ahead	one period, stored	one period, differentiated	one period, computed	one period, target net

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- **Same letters: the OLG versions differ in what a datum is and how long it lives.**

Source: Deck 4.2, *Three Pipelines, the Same Letters*; the two symbol pages of this deck.

Bridge: Each Tool for Its Question

Paths: one unanticipated shock, any distribution.

The state as input: when Γ is a short vector, as here.

The history as input: when Γ is many distributions.

Next: back to Deck 4.1's histogram, and a government that taxes on it.

Source: The sources of this deck.

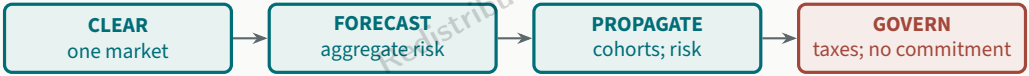
- Age turns Γ into a stack. Without ε the stack is a vector of $2J$ numbers, and Deep Equilibrium Nets solve it exactly.
- With ε the stack is many histograms: the history is the input, continuation values the output, EGM the solver.
- **Deck 4.4: a government that cannot commit chooses a tax rule on the whole distribution.**

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Session 4.4

Optimal Taxation Without Commitment

Put a government into Deck 4.1's economy: a tax rule on the whole distribution, a Markov-perfect fixed point in rules, and the algorithm that finds it

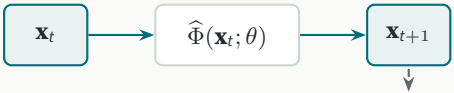


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Where Decks 4.1 and 4.2 Stopped



Deck 4.1, a household: $\mathbf{x}_{t+1}$ taken as given



this deck, a government: it sees $d\mathbf{x}_{t+1}$

- Deck 4.1: the histogram $\mathbf{x}$ and its matrix $\widehat{\Phi}$, rebuilt at every date inside the graph. A household is atomistic: it takes $\mathbf{x}_{t+1}$ as given.
- Deck 4.2, *Argument 1*: a government that chooses its tax every period needs $d\Gamma'/d\tau$, and only a law of motion in the graph supplies it. Here τ is a function of $\mathbf{x}$, so households respond to it through $\mathbf{x}$.
- This deck puts that government into Deck 4.1's economy.
- **One object is the state, the law of motion and the gradient's path.**

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Source: Feng, Han, Sargent & Zhu (September 2026), preliminary; Decks 4.1 and 4.2.

The Question: a Tax Nobody Can Promise



- Under commitment the long-run capital tax is zero in the classic models: promise not to tax capital, and people save.
- Capital in place is an inelastic base. A government that chooses afresh each period is tempted to tax it; savers know it.
- Without commitment the problem is a game between today's government and its successors.
- The answer is a rule on the whole distribution, $\tau(\Gamma)$: not a number, not a function of a few moments.
- **Sought: the tax a government chooses when every successor will choose by the same rule.**

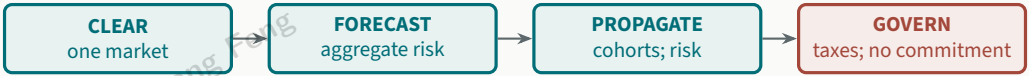
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Source: Feng, Han, Sargent & Zhu (September 2026), preliminary; Chamley (1986); Judd (1985); Klein, Krusell & Ríos-Rull, *RES* (2008).

Session 4.4

The Economy and Its Equilibrium

Deck 4.1's households, a government that taxes and cannot commit, and a fixed point in rules



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Households

- As in Deck 4.1: ε redrawn every period (i.i.d.), $Q = 5$ Gauss–Hermite nodes (ε_q, p_q), r net of depreciation. A household maximises

$$\sum_{t=0}^{\infty} \beta^t \, U(c_t, l_t, G_t), \qquad U(c, l, G) = u\Big(c - \psi_l \frac{l^{1+\theta_l}}{1+\theta_l}\Big) + \gamma_g \log G, \qquad u(y) = \frac{y^{1-\sigma}}{1-\sigma},$$

  subject to

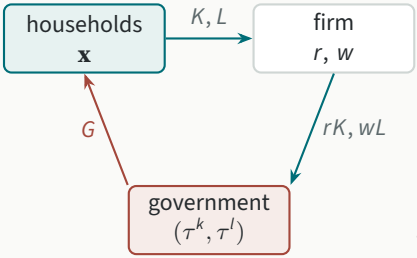
$$c_t + a_{t+1} = \big[1 + (1-\tau_t^k)\, r_t\big]\, a_t + (1-\tau_t^l)\, w_t\, \varepsilon_t\, l_t, \qquad a_{t+1} \geq 0.$$

- τ^k taxes capital income, τ^l labour income; G is a public good, valued with weight γ_g.
- G enters additively, so it changes no household's choice: only the taxes do.
- **Hours still carry no wealth effect: the Frisch elasticity is $1/\theta_l = 3.125$.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary. $\beta = 0.95$, $\sigma = 2$, $\theta_l = 0.32$, $\psi_l = 1.5$, $\sigma_\varepsilon = 0.2$.

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Firms, a Balanced Budget, Market Clearing



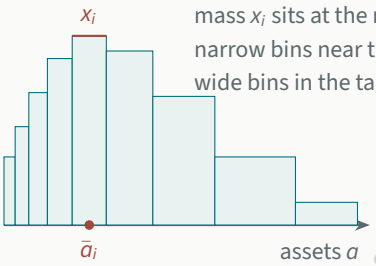
no bonds: G is whatever the taxes raise this period

- Firm: $Y = K^\alpha L^{1-\alpha}$, $r = \alpha(K/L)^{\alpha-1} - \delta$, $w = (1-\alpha)(K/L)^\alpha$.
- Government: picks (τ^k, τ^l) and balances its budget every period, $G = \tau^k rK + \tau^l wL$.
- Markets: $K = \sum_i x_i \bar{a}_i$, $L = \mathbb{E}[\varepsilon\, l(\varepsilon)]$, exactly; the histogram moves by $\mathbf{x}_{t+1} = \mathbf{x}_t \widehat{\Phi}_t$.
- No bonds and no commitment technology: today's government cannot bind its successors.
- G **is a residual, not a choice: two instruments, one budget, every period.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary. $\alpha = 0.34$, $\delta = 0.06$.

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The Distribution, Approximated by a Histogram



mass x_i sits at the midpoint $\bar{a}_i$
narrow bins near the limit,
wide bins in the tail

- $N = 140$ bins on $[0, 16]$, midpoints $\bar{a}_1 < \cdots < \bar{a}_N$; the state is the row vector $\mathbf{x} = (x_1, \ldots, x_N)$, $x_i \geq 0$, $\sum_i x_i = 1$.
- Aggregates are sums: $K = \sum_i x_i \bar{a}_i$.
- ε is i.i.d.: $Q = 5$ Gauss–Hermite nodes (ε_q, p_q) integrate it, so $\mathbf{x}$ needs no ε dimension.
- **The government's state is these N numbers.**

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Source: Feng, Han, Sargent & Zhu (September 2026), preliminary; as in Deck 4.1. The grid is fixed once, from the classical benchmark.

The Law of Motion $\mathbf{x}_{t+1} = \mathbf{x}_t \widehat{\Phi}_t$, Built Entry by Entry

At date t, for every source bin i and node q

1. saving at the midpoint: $a'_{t,iq} = g_\theta(\varepsilon_q, \bar{a}_i; \mathbf{x}_t)$, kept inside $[\bar{a}_1, \bar{a}_N]$
2. bracket it, $\bar{a}_j \le a'_{t,iq} \le \bar{a}_{j+1}$ with $j = j_{t,iq}$, and weigh the upper neighbour: $\omega_{t,iq} = \dfrac{a'_{t,iq} - \bar{a}_j}{\bar{a}_{j+1} - \bar{a}_j}$
3. send $p_q(1 - \omega_{t,iq})$ of bin i's mass to bin j and $p_q\,\omega_{t,iq}$ to bin $j+1$ (Young's lottery)

$$\widehat{\Phi}_{t,ij'} = \sum_{q=1}^{Q} p_q \Big[(1 - \omega_{t,iq})\, \mathbf{1}\{j' = j_{t,iq}\} + \omega_{t,iq}\, \mathbf{1}\{j' = j_{t,iq} + 1\} \Big], \qquad x_{t+1,j'} = \sum_{i=1}^{N} x_{t,i}\, \widehat{\Phi}_{t,ij'}.$$

- Row i says where bin i's households go: rows sum to one, so mass is conserved.
- Exact in the mean: $\sum_{j'} \widehat{\Phi}_{t,ij'} \bar{a}_{j'} = \sum_q p_q a'_{t,iq}$, so K_{t+1} equals what households save.
- $\widehat{\Phi}_t$ **moves with θ and with $\mathbf{x}_t$, through a': it is rebuilt at every date, and ω is linear in a', so it can be differentiated.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary; Young, *JEDC* (2010); Deck 4.1. $N \times Q$ network calls per date; no simulation, no sampling noise.

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Household Conditions

- Saving: the Euler condition with the limit, equality whenever $a' > 0$,

$$u'(y_t) \;\geq\; \beta\,\mathbb{E}_{\varepsilon'}\Big\{\big[1+(1-\tau^k_{t+1})\,r_{t+1}\big]\,u'(y_{t+1})\Big\}, \qquad y = c - \psi_l \frac{l^{1+\theta_l}}{1+\theta_l}.$$

- Hours, in closed form:

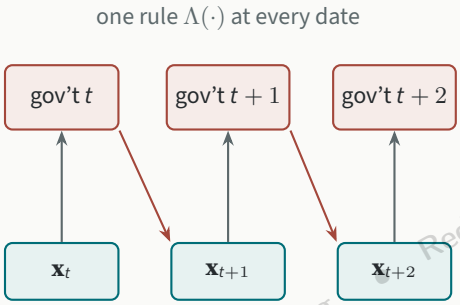
$$(1-\tau^l)\,w\,\varepsilon = \psi_l\, l^{\theta_l} \quad\Longrightarrow\quad l(\varepsilon) = \Big(\frac{(1-\tau^l)\,w\,\varepsilon}{\psi_l}\Big)^{1/\theta_l}.$$

- *Tomorrow's* capital tax sits inside *today's* saving decision: expectations about successors move the economy now.
- Hours contain no a. Given τ^l and K, labour L, both prices and both tax bases are fixed, whatever the shape of $\mathbf{x}$.
- **Under GHH the marginal utility of consumption is $u'(y)$, so $U_c = u'(y)$ in every condition below.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary; Greenwood, Hercowitz & Huffman, *AER* (1988).

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State, Rule, Timing



- State: $(\varepsilon, a; \mathbf{x})$ for a household; $\mathbf{x}$ alone for the government.
- A **Markov rule** $\Lambda(\mathbf{x}) = \left(\tau^k(\mathbf{x}), \tau^l(\mathbf{x})\right)$, with $G(\mathbf{x})$ from the budget: no calendar time, no history.
- Every government, at every date, uses the same rule, and that rule is the unknown.
- Households take a perceived rule $\tilde{\Lambda}$ and law of motion Φ as given, and best-respond.
- **The Markov restriction, not foresight, is what makes the rule time-consistent.**

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Source: Feng, Han, Sargent & Zhu (September 2026), preliminary; Klein, Krusell & Ríos-Rull (2008).

The Two Problems

(P-1) Household. Given prices, G, the perceived rule $\tilde{\Lambda}$ and $\mathbf{x}_+ = \mathbf{x}\widehat{\Phi}$,

$$V(\varepsilon, a; \mathbf{x}) = \max_{0 \le a'} \Big\{ U(c, l, G) + \beta \sum_{q'} p_{q'} \, V(\varepsilon_{q'}, a'; \mathbf{x}_+) \Big\}.$$

(P-2) Government (Deck 4.2's V^G). Given the successors' rule $\tilde{\Lambda}$ and its value $\widetilde{W}$,

$$W(\mathbf{x}) = \max_{\tau^k, \tau^l} \Big\{ \sum_{i,q} x_i \, p_q \, U(c_{iq}, l_q, G) + \beta \, \widetilde{W}(\mathbf{x}_+) \Big\},$$

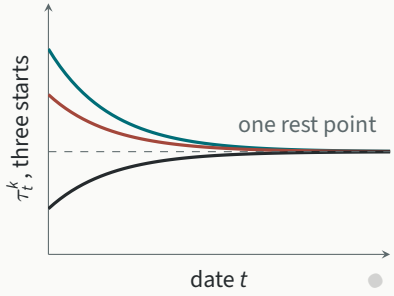
subject to the budget, prices, the household's conditions at *every* node $(\bar{a}_i, \varepsilon_q)$, and $\mathbf{x}_+ = \mathbf{x}\widehat{\Phi}$.

- Welfare is utilitarian: each node counts by its mass $x_i p_q$.
- **The asymmetry: the household takes $\mathbf{x}_+$ as given; the government is not atomistic and sees how its choice moves $\mathbf{x}_+$.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary; Klein, Krusell & Ríos-Rull (2008).

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Markov-Perfect Equilibrium



- A **Markov-perfect equilibrium**: saving and value functions, prices, a rule Λ with value W, and perceptions, with

  (a) households solving (P-1); (b) $\Lambda(\mathbf{x})$ solving (P-2) at every $\mathbf{x}$; (c) perceived = actual: $\tilde{\Lambda} = \Lambda$, $\widetilde{W} = W$, Φ from households' saving; (d) markets clearing.
- No aggregate shocks: from *any* $\mathbf{x}_0$, taxes and aggregates must settle to constants.
- **A sharp test: paths from different starts must meet.**

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Source: Feng, Han, Sargent & Zhu (September 2026), preliminary. Schematic curves: direction and shape only.

Three Computational Challenges

Challenge	Deck 4.2's methods	Here
the distribution is a state	learned moments of a panel; a histogram; a history of shocks	Deck 4.1's histogram, and quadrature for ε
inequality constraints	Fischer–Burmeister: two inequalities become one equation	the same, at every node, priced by a penalty
strategic interaction	none of the three has a government that re-chooses	Deck 4.1's matrix, with the gradient passed through it

- Simulating Γ_+ works when nobody acts on the distribution; with a player who does, it breaks the forward and backward passes (Deck 4.2, *Argument 2*).
- **Deck 4.1 settled the first row. This deck takes the other two, then asks: is the answer right?**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary; Maliar et al. (2021); Azinovic et al. (2022); Han, Yang & E (2026).

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Implementability: One Equation per Household State

- At every node $(\bar{a}_i, \varepsilon_q)$ the government must respect the Euler condition with the limit, as one equation:

$$\mathrm{FB}(\vartheta_1, \vartheta_2) = \vartheta_1 + \vartheta_2 - \sqrt{\vartheta_1^2 + \vartheta_2^2 + \eta^2} = 0, \qquad \bar{E}_{iq} = \mathrm{FB}_{iq} / u'(y_{iq}),$$

$$\vartheta_1 = u'(y) - \beta \sum_{q'} p_{qq'} \left[1 + (1 - \bar{\tau}^k(\mathbf{x}_+))\, r(\mathbf{x}_+)\right] u'(y_{+,q'}), \qquad \vartheta_2 = a'.$$

- $\mathrm{FB} = 0$ when both are non-negative and one is zero; $\eta = 10^{-4}$. $\bar{E}$ is a relative Euler error.
- a' is the saving *after* the clamp, as the budget and $\widehat{\Phi}$ use it (Deck 4.1: pre-clamp $\tilde{a}$).
- Tomorrow's tax is the *perceived* rule $\bar{\tau}^k$, at the $\mathbf{x}_+$ today's saving produces.
- **The function is held fixed; the state it is evaluated at is not. There the strategic channel lives.**

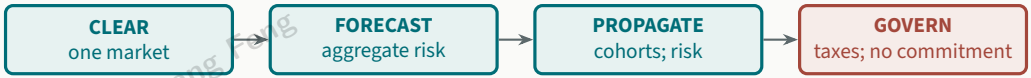
Source: Feng, Han, Sargent & Zhu (September 2026), preliminary; Maliar, Maliar & Winant (2021).

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Session 4.4

The Ingredients and the Algorithm

Three networks, one penalised objective, a rollout through the matrix, and a frozen copy of the future



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Three Trained Networks

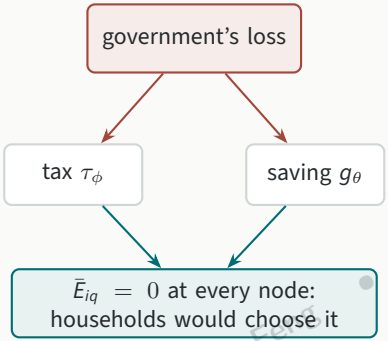
Network	Input	Out	Head
saving g_θ	$(\varepsilon, a, \mathbf{x})$	1	Deck 4.1's change head, $\tilde{a} = a + D(2s-1)$, $a' = \min\{\max\{\tilde{a}, 0\}, \bar{a}\}$
tax rule τ_ϕ	$\mathbf{x}$	2	a sigmoid times a cap: $\tau^k = \tau^k_{\max}\sigma(\cdot)$, $\tau^l = \tau^l_{\max}\sigma(\cdot)$
aggregate value $\bar{V}_\chi$	$\mathbf{x}$	1	linear: the government's value beyond the rollout
household critic V_ψ	$(\varepsilon, a, \mathbf{x})$	1	Deck 4.1's critic: loaded, not trained here

- Each: two hidden layers of 256 rectified-linear units. The tax rule reads the whole histogram, so it can respond to its *shape*.
- **The saving network reads $\mathbf{x}$, not the tax: in equilibrium the tax is itself a function of $\mathbf{x}$.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary. $D = 1.3$, as in Deck 4.1.

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The Government Chooses the Allocation Too

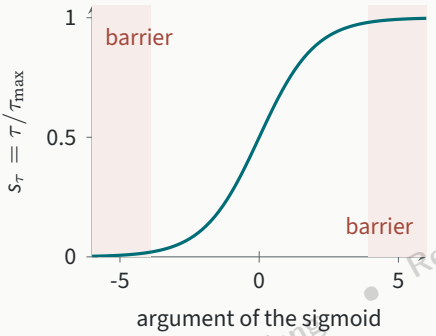


- Deck 4.1 trained saving on the household's own Bellman equation. Here it learns only from the government's loss.
- The government picks the tax *and* the saving rule; $\bar{E} = 0$ makes it one households would choose.
- The **primal approach** of Ramsey taxation, with a Markov government in place of a committed one.
- **The rule is the households' only as far as the penalty drives $\bar{E}$ to zero.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary; Chari & Kehoe, *Handbook of Macroeconomics* (1999), ch. 26.

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The Tax Rule's Head Decides What Can Be Learned

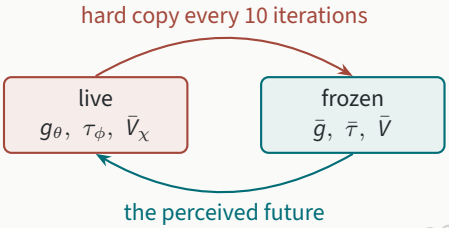


- $\tau^k(\mathbf{x}) = \tau^k_{\max} s_\tau$, $s_\tau = \sigma(\ell(\mathbf{x}))$, ℓ the last layer: its slope carries $\tau^k_{\max} s_\tau(1 - s_\tau)$.
- At the cap, $s_\tau \to 1$ and the gradient vanishes. A constant tax is a trivial fixed point: the run looks converged.
- Remedy one: a cap $\tau_{\max}$ wide enough for an interior optimum. A solver setting, not economics.
- Remedy two: a hinge barrier, zero inside, $B_\sigma = \sum \left[(0.02 - s_\tau)^2_+ + (s_\tau - 0.98)^2_+\right]$.
- **A floor under the gradient wherever τ lands, or the rule cannot learn to respond to $\mathbf{x}$.**

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Source: Feng, Han, Sargent & Zhu (September 2026), preliminary.

Perceived Equals Actual: the Frozen Copy



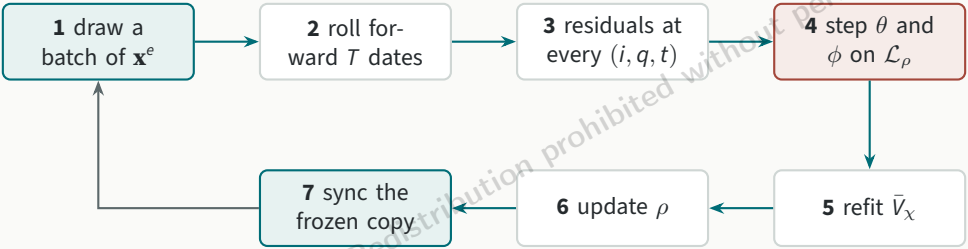
today's choices are trained; tomorrow's rule, saving and value are read from the copy

- The frozen copy supplies $\bar{\tau}^k(\mathbf{x}_+)$ and $\bar{g}$ inside ϑ_1, and $\bar{V}(\mathbf{x}_T)$ at the end of the rollout.
- Inside one step these perceptions do not move: they are targets, not parameters.
- Copying the live networks into the frozen ones imposes consistency: perceived $=$ actual.
- It is Session 3's target network, now used to reach a fixed point in rules.
- **At convergence the rule the government expects of its successors is the rule it plays.**

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Source: Feng, Han, Sargent & Zhu (September 2026), preliminary; Session 3, *Bridge (3): the Household's Actor–Critic*.

The Loop in One Picture

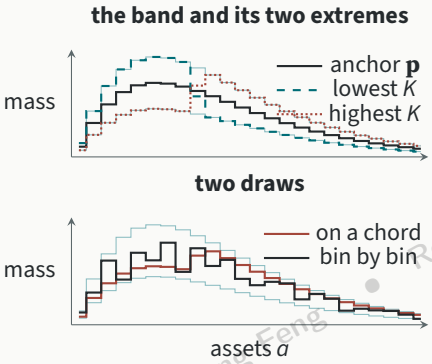


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- Steps 1–4 run four times per iteration, each on a fresh batch; steps 5–7 once, or at longer intervals.
- Steps 2–4 are one forward and one backward pass through $T = 30$ matrices.
- **No outer loop: the fixed point in rules is reached by training and synchronising.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary.

Step 1a: Sampling Distributions From a Band

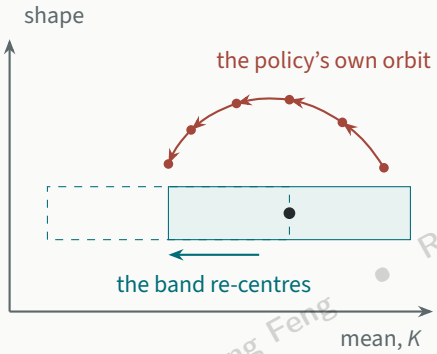


- Around an anchor $\mathbf{p}$, each bin may move by a share ν, $p_i(1-\nu) \le x_i \le p_i(1+\nu)$ (thin lines); the total stays one.
- **Chords**: free mass to the richest, or poorest, bins gives the highest, or lowest, K. Draw K uniformly on the line from $\mathbf{p}$ to that extreme.
- **Bin by bin**: each x_i uniform in its interval, projected back to mass one: every shape, little range in K.
- **Chords give reach in K, bin-by-bin draws give shape: the rule learns only where it is sampled.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary. Schematic: an illustrative anchor on 24 bins, $\nu = 0.35$.

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Step 1b: The Band Follows, and the Policy's Own Orbit



- $\mathbf{x}^e, e = 1, \ldots, E$: a batch of starting histograms, one rollout each. Half come from the band (Step 1a).
- Anchor $p_i \propto x_i^* e^{\kappa \bar{a}_i}$, moved every 10 iterations halfway to the classical K^* at current taxes: a taxing government moves both.
- Half from an **orbit**: every 20 iterations, $\mathbf{x} \leftarrow \mathbf{x}\widehat{\Phi}(\mathbf{x}; \theta)$ for 60 steps, no gradient, plus the rule's own stationary $\mathbf{x}$.
- **The band covers off the path; the orbit is accurate on it.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary.

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Step 2: Roll Forward T Dates

From one sampled $\mathbf{x}_0 = \mathbf{x}^e$, for $t = 0, \ldots, T-1$

1. $K_t = \sum_i x_{t,i} \bar{a}_i$; taxes $(\tau_t^k, \tau_t^l) = \tau_\phi(\mathbf{x}_t)$; hours l_q, then L_t, r_t, w_t: no price loop
2. saving $a'_{t,iq} = g_\theta(\varepsilon_q, \bar{a}_i; \mathbf{x}_t)$; consumption $c_{t,iq}$ from the budget; $y_{t,iq}$
3. revenue $G_t = \sum_{i,q} x_{t,i} p_q \left[\tau_t^k r_t \bar{a}_i + \tau_t^l w_t \varepsilon_q l_q\right]$
4. flow welfare $\bar{U}_t = \sum_{i,q} x_{t,i}\, p_q \left[u(y_{t,iq}) + \gamma_g \log(G_t + 10^{-3})\right]$
5. build $\widehat{\Phi}_t$ from the $a'_{t,iq}$ by Deck 4.1's lottery; $\mathbf{x}_{t+1} = \mathbf{x}_t \widehat{\Phi}_t$, *kept in the graph*

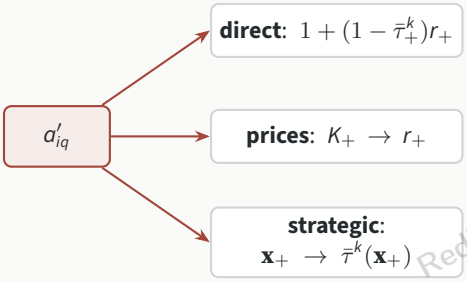
$$W(\mathbf{x}^e; \theta, \phi) = \sum_{t=0}^{T-1} \beta^t \bar{U}_t + \beta^T \bar{V}(\mathbf{x}_T), \qquad \bar{V} \text{ the frozen copy, a constant.}$$

- **Deck 4.1's rollout, with a tax at every date, the public good in welfare, and the matrix left in the graph.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary. $T = 30$; utility floored far below any value reached, to guard the logarithm and the power.

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Step 3: The Constraint at Every (i, q, t), and Its Three Channels



each moves ϑ_1; all three stay in the graph

- At each (t, i, q) and batch member: $\bar{E}_{t,iq} = \mathrm{FB}(\vartheta_1, \vartheta_2)/u'(y_{t,iq})$.
- Inside ϑ_1: $\mathbf{x}_+$ from the *live* saving rule; $\bar{\tau}$ and $\bar{g}$ from the frozen copy at $\mathbf{x}_+$.
- Deck 4.1 held a household's continuation fixed: it is its own marginal value (the envelope theorem).
- Here ϑ_1 is a *constraint*: no envelope theorem, and dropping a channel stalls the penalty.
- **The strategic channel, $\mathbf{x}_+ \to \bar{\tau}^k(\mathbf{x}_+)$, is the one no household sees.**

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Source: Feng, Han, Sargent & Zhu (September 2026), preliminary.

Step 4: The Government's Loss

$$\mathcal{L}_\rho(\theta,\phi) = -W + \sum_{t,i,q} \beta^t x_{t,i} p_q \left[\mu_{t,iq}\bar{E}_{t,iq} + \frac{\rho}{2}\bar{E}^2_{t,iq}\right] + B_{\text{feas}} + B_\sigma(\mathbf{x}^e) + \kappa_{\text{band}} B_{\text{band}}$$

- An augmented Lagrangian, multipliers μ. **As run, $\mu \equiv 0$: a pure quadratic penalty, weight ρ.**
- Weights $\beta^t x_{t,i} p_q$ price each residual by its mass; B_{feas}, same weights: $\left[(-c)_+ + (-y)_+\right]/10^{-4}$.
- $B_{\text{band}} = \sum_t \beta^t \sum_i \left[(p_i(1-\nu) - x_{t,i})_+ + (x_{t,i} - p_i(1+\nu))_+\right]$, $\kappa_{\text{band}} = 20$: rollouts stay in the trained region.
- $B_\sigma = \sum_\tau \left[(0.02 - s_\tau)_+^2 + (s_\tau - 0.98)_+^2\right]$, $s_\tau = \tau/\tau_{\max}$, at the sampled $\mathbf{x}^e$: zero inside, it keeps the sigmoid's slope alive, so the rule cannot freeze at its cap.
- No gradient through the weights, the $u'(y)$ dividing $\bar{E}$, or $\bar{V}(\mathbf{x}_T)$; one backward pass, one Adam step on θ and one on ϕ.
- **Every term but $-W$ is a solver device; the economics is $-W$ and $\bar{E} = 0$.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary; Nocedal & Wright, *Numerical Optimization* (2006), ch. 17.

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How a Network Responds to the Distribution

- The N bin masses enter the first layer like any other input. For the tax rule, two hidden ReLU layers:

$$\tau_\phi^k(\mathbf{x}) = \tau_{\max}^k \, \sigma\big(\ell_\phi(\mathbf{x})\big), \qquad \ell_\phi(\mathbf{x}) = w_3^\top \mathrm{ReLU}\big(W_2 \, \mathrm{ReLU}(W_1 \mathbf{x}^\top + b_1) + b_2\big) + b_3.$$

- Its response to the mass in bin i, with D_1, D_2 the diagonal masks of active units:

$$\frac{\partial \tau_\phi^k}{\partial x_i} = \tau_{\max}^k \, \sigma'(\ell_\phi) \, \left[w_3^\top D_2 W_2 D_1 W_1\right]_i.$$

- Mass sums to one, so only moves count: δ from bin j to bin i changes τ^k by $\delta\,(\partial_i \tau^k - \partial_j \tau^k)$. A rise in K is one such direction; a mean-preserving spread is another.
- The saving rule $g_\theta(\varepsilon, \bar{a}_i; \mathbf{x})$ and the value $\bar{V}_\chi(\mathbf{x})$ read $\mathbf{x}$ the same way.
- $\partial\tau/\partial\mathbf{x}$ **is learned, not imposed: it is how the government answers inequality, and it enters every saving update.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary. Two hidden layers of 256 units; τ^l has its own output ℓ.

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Updating the Tax Rule, One Date at a Time

$$\nabla_\phi \mathcal{L}_\rho = \sum_{t=0}^{T-1} \left[\frac{\partial \mathcal{L}_\rho}{\partial \tau_t^k} \nabla_\phi \tau_\phi^k(\mathbf{x}_t) + \frac{\partial \mathcal{L}_\rho}{\partial \tau_t^l} \nabla_\phi \tau_\phi^l(\mathbf{x}_t) \right] + \nabla_\phi B_\sigma,$$

$$\frac{\partial \mathcal{L}_\rho}{\partial \tau_t^k} = -\beta^t r_t \Big[\underbrace{\frac{\gamma_g K_t}{G_t + 10^{-3}}}_{\text{value of revenue}} - \underbrace{\sum_{i,q} x_{t,i} p_q \, u'(y_{t,iq}) \, \bar{a}_i}_{\text{cost to households}} \Big] + \rho \sum_{i,q} \beta^t x_{t,i} p_q \, \bar{E}_{t,iq} \, \frac{\partial \bar{E}_{t,iq}}{\partial \tau_t^k}.$$

- g_θ does not read τ, so $\mathbf{x}_t$ does not depend on ϕ: each term sits at one date, at the $\mathbf{x}_t$ reached.
- The bracket is the flow trade-off, node by node; the penalty enters through today's $u'(y)$ in ϑ_1.
- **Adam pushes τ_ϕ toward this condition at every $\mathbf{x}$ visited: the rule learns its response to the distribution only where the sampler and rollouts take it.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary. r net; $\partial \bar{E} / \partial \tau_t^k$ through $u'(y_{t,iq})$, the divisor held fixed. τ^l: the same form, plus terms through hours, L, r, w.

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Updating the Saving Rule: Through the Law of Motion

$$\mathbf{x}_{t+1} = \mathbf{x}_t\,\widehat{\Phi}_t, \qquad \widehat{\Phi}_t = \widehat{\Phi}\big(a'_t\big), \qquad a'_{t,iq} = g_\theta(\varepsilon_q, \bar{a}_i; \mathbf{x}_t), \qquad \frac{d\mathbf{x}_0}{d\theta} = 0.$$

$$\frac{d\mathbf{x}_{t+1}}{d\theta} = \underbrace{\frac{d\mathbf{x}_t}{d\theta}\,\widehat{\Phi}_t}_{\text{displaced mass, carried on}} + \underbrace{\mathbf{x}_t\,\frac{\partial \widehat{\Phi}_t}{\partial \theta}}_{\text{where today's mass lands}} + \underbrace{\mathbf{x}_t\,\frac{\partial \widehat{\Phi}_t}{\partial \mathbf{x}_t}\,\frac{d\mathbf{x}_t}{d\theta}}_{\text{the saving rule reads } \mathbf{x}_t}$$

- Product rule on $\mathbf{x}_t\widehat{\Phi}_t$, then chain rule: $\widehat{\Phi}_t$ moves with θ directly and through $\mathbf{x}_t$.

Parameters	**The gradient reaches them through**
saving θ	consumption; the three terms, date after date; the taxes at each $\mathbf{x}_t$; the channels of ϑ_1
tax rule ϕ	the budget; G in welfare; $u'(y)$ in $\bar{E}$. Never $\mathbf{x}_+$: g_θ does not read τ
neither	$\bar{V}(\mathbf{x}_\tau)$: the frozen copy, a constant

- **Through θ, the gradient runs back along every $\widehat{\Phi}_t$; through ϕ, only within the date.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary. $\partial\widehat{\Phi}/\partial a'$ is the lottery's slope, $\pm 1/(\bar{a}_{j+1} - \bar{a}_j)$, zero where the clamp binds; the third term contracts over $\mathbf{x}_t$. Autograd runs the same chain in reverse.

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Updating the Saving Rule: What the Distribution Sends Back

$$\boldsymbol{\lambda}_t \equiv \frac{d\mathcal{L}_\rho}{d\mathbf{x}_t} = \frac{\partial \ell_t}{\partial \mathbf{x}_t} + \Big(\boldsymbol{\lambda}_{t+1} + \frac{\partial \ell_t}{\partial \mathbf{x}_{t+1}}\Big) J_t, \quad J_t = \frac{\partial \mathbf{x}_{t+1}}{\partial \mathbf{x}_t}, \quad \boldsymbol{\lambda}_T = 0,$$

$$\nabla_\theta \mathcal{L}_\rho = \sum_{t=0}^{T-1} \Big[\frac{\partial \ell_t}{\partial \theta} + \Big(\boldsymbol{\lambda}_{t+1} + \frac{\partial \ell_t}{\partial \mathbf{x}_{t+1}}\Big) \mathbf{x}_t \frac{\partial \widehat{\Phi}_t}{\partial \theta}\Big], \qquad \ell_t : \text{ date } t\text{'s terms of } \mathcal{L}_\rho.$$

In $\partial \ell_t / \partial \mathbf{x}_t$	who sits where, $\sum_q p_q U_{iq}$; prices via K_t; **the tax rule's answer** $\partial \tau_\phi / \partial \mathbf{x}_t$; g_θ's input
In $\partial \ell_t / \partial \mathbf{x}_{t+1}$	in $\bar{E}_t$: $\bar{\tau}^k(\mathbf{x}_{t+1})$, $r(\mathbf{x}_{t+1})$, $\bar{g}(\cdot; \mathbf{x}_{t+1})$: the strategic channel

- The tax rule's answer reaches θ twice: through $\tau_\phi(\mathbf{x}_t)$ along the path, and through $\bar{\tau}(\mathbf{x}_{t+1})$ in each Euler residual.
- **The saving rule is updated knowing how the tax will answer the distribution its own saving creates.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary. The previous page run in reverse: one backward pass, $d\mathbf{x}_t / d\theta$ never formed. Penalty weights held fixed: "who sits where" enters through welfare only; $\bar{V}(\mathbf{x}_T)$ sends nothing back.

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What Is New Here

	the state	who acts on $\mathbf{x}$	the gradient through $\mathbf{x}$'s motion
Krusell–Smith; Deck 4.2	mean, moments, histogram, shocks	nobody: households take it as given	one period, or a household panel
Markov taxes, one agent	K	a government	no distribution
HA taxes, commitment	a given $\mathbf{x}_0$	a government, once	a path of taxes, not a rule
this deck	**the whole histogram**	**a government, every date**	**T dates, with $\partial\tau/\partial\mathbf{x}$ inside**

- Without commitment the tax is a function $\tau(\mathbf{x})$; with commitment, a path chosen once.
- **To the authors' knowledge, the first time-consistent tax rule computed on the whole distribution as the state.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary. One agent: Klein, Krusell & Ríos-Rull (2008). Commitment: Bhandari et al. (2021); Dyrda & Pedroni (2022).

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Steps 5 and 7: Refit the Value Beyond the Horizon; Sync

- **Targets**, every 20 iterations, at fresh band draws $\mathbf{x}$. As run, flow welfare held forever:

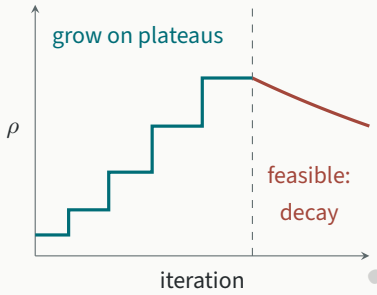
$$\hat{V}(\mathbf{x}) = \frac{\bar{U}(\mathbf{x})}{1-\beta}, \qquad \bar{U}(\mathbf{x}) = \sum_{i,q} x_i \, p_q \, U_{iq}.$$

- **Fit**, every iteration: 20 Adam steps on $\text{mean}[\bar{V}_\chi(\mathbf{x}) - \hat{V}(\mathbf{x})]^2$, batches of 256, after a long prefit.
- **Sync**, every 10 iterations: $g_\theta, \tau_\phi, \bar{V}_\chi$ copied into $\bar{g}, \bar{\tau}, \bar{V}$. A hard copy, not a moving average.
- Beyond T the economy is valued as if it stayed where the rollout ended; at $\beta^T \approx 0.2$ that term is not negligible.
- $\bar{V}$ **supplies a level at the horizon, never a gradient; the rollout supplies the gradient.**

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Source: Feng, Han, Sargent & Zhu (September 2026), preliminary.

Step 6: The Penalty Weight Follows the Residual



- Each iteration, read $\bar{E}^{\text{test}}$: the mass-weighted mean of $|\bar{E}|$ on a fixed test batch.
- Grow, $\rho \leftarrow \min(1.015\rho, 10^4)$, if the loss stalls ($|\Delta\mathcal{L}/\mathcal{L}| \leq 1\%$) while $\bar{E}^{\text{test}} \geq$ tol and not improving.
- Decay, $\rho \leftarrow 0.99\rho$, once 5 of the last 10 reads are below tol.
- Learning rates scaled by $1/\sqrt{1+0.1\rho}$.
- **Measure tol at the solution: below what is reachable, ρ grows without end.**

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Source: Feng, Han, Sargent & Zhu (September 2026), preliminary. Schematic curves: direction and shape only.

The Algorithm

Taxation without commitment, as a trained fixed point

1 warm-start g_θ from the no-tax solve; tax rule near zero; prefit $\bar{V}_\chi$; copy all to the frozen set
2 four times: draw $\mathbf{x}^e$, half band, half orbit (step 1)
3 roll forward T dates, $\widehat{\Phi}_t$ in the graph; form W and every $\widehat{E}_{t,iq}$ (steps 2–3)
4 one backward pass of $\mathcal{L}_\rho$; one Adam step on θ, one on ϕ (step 4)
5 refit $\bar{V}_\chi$; every 20 iterations new targets and orbit, every 10 a moved anchor (step 5)
6 update ρ and the learning rates from the test-batch residual (step 6)
7 every 10 iterations: frozen $\leftarrow$ live, "perceived $=$ actual" (step 7)
8 stop at the iteration budget, or early if the Euler error blows up; keep the best checkpoint

- **Each line is Deck 4.1's machinery or Session 3's target network. What is new is who owns the loss.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary.

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What the Gradient Requires

Requirement	How it is met
a differentiable transition	lottery weights piecewise linear in a' (Deck 4.1); kinks of measure zero
no sampling noise	quadrature over ε, every asset node: a deterministic loss
a gradient in the tax rule	the barrier keeps the sigmoid off 0 and 1
a gradient at the limit	$\vartheta_2 = a'$ and the smoothed root keep constrained nodes live
memory for the graph	T dates of $N \times Q$ evaluations, $N \times N$ matrices, next-date copies in ϑ_1

- T is a budget. Beyond it, $\bar{V}$ adds a level and no gradient.
- Lab block B: derivatives through up to twenty lottery steps match finite differences to 2×10^{-10}.
- **Every row is a place where a gradient can vanish without an error message.**

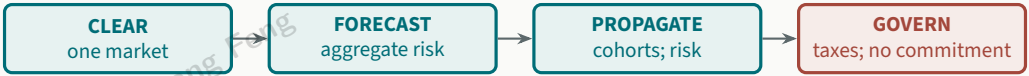
Source: Feng, Han, Sargent & Zhu (September 2026), preliminary; the lab, block B.

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Session 4.4

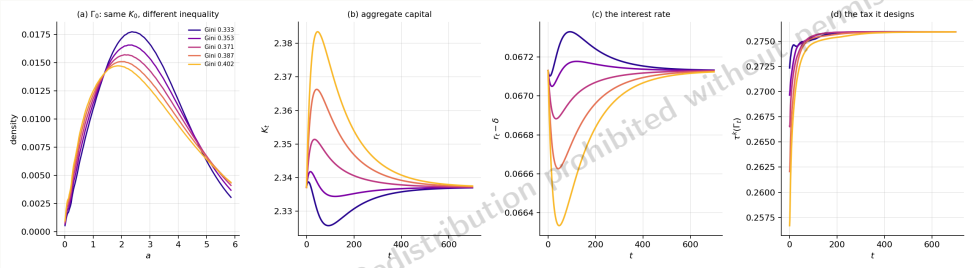
What the Rule Does Along a Path

Start the trained economy from different distributions and let the tax rule, saving and prices run



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At the Same Capital, Inequality Alone Moves the Tax



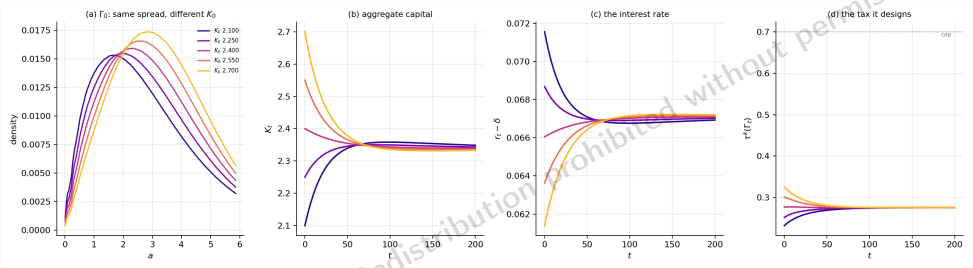
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- Five starts Γ_0 share one K_0 and differ only in inequality (a). At date 0, hours, prices and both tax bases are identical.
- The more unequal start is taxed less on impact (d); its capital overshoots further before returning (b, c).
- Every start converges to the same Γ^* and the same long-run tax.
- **A property of the trained rule: why it responds to shape is not yet settled.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary. Simulated with the trained Stage 2A networks.

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At the Same Inequality, More Capital Means a Bigger Opening Levy



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- The mirror experiment: the spread is held and K_0 moves (a). Now the tax base moves too, and the tax responds far more (d).
- The levy is transitional: capital and the interest rate return (b, c), and every start reaches the same long-run rate.
- A cap that never binds at the rest point can still censor the path: size it for the levy.
- **Without commitment: tax the capital inherited, then settle down.**

Source: Feng, Han, Sargent & Zhu (September 2026), preliminary. Simulated with the trained Stage 2A networks.

Bridge: the Lab, and Session 5

Lab, Part I: the classical benchmarks of Decks 4.1–4.3.

Blocks D, E, F: Krusell–Smith by DeepHAM, by Deep Equilibrium Nets

Blocks G, H: the OLG with a market-clearing layer and a homotopy; the retirement OLG in the sequence space.

This deck: a government that differentiates through Γ.

- Cost: GPU-hours, not seconds. Stage 1B alone took about 2.9 GPU-hours, against 30 CPU-seconds for Stage 1A.
- It buys rules on the whole distribution, and the path from any start in the sampled region.
- Next in the research: aggregate shocks, $\widehat{\Phi}$ conditioned on (z, z'); then persistent ε.
- **The lab trains the Deck 4.2 and 4.3 networks in PyTorch, each graded by a gate fixed before the run.**

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Source: Feng, Han, Sargent & Zhu (September 2026), preliminary; the Session 4 lab.